

IUS
INSTITUT
UNIVERSITAIRE
DES SCIENCES

FACULTE DES SCIENCES ET TECHNOLOGIE (FST)

- **Cours : RESEAU I**
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- **Professeur : Ismael SAINT AMOUR**

LE

INTRODUCTION

Dans le cadre du cours Réseaux I, ce troisième travail dirigé porte sur l'adressage IPv4 et IPv6 à l'aide du logiciel Cisco Packet Tracer.

Ce travail pratique nous permet de comprendre comment attribuer des adresses IP aux différents équipements d'un réseau informatique et comment vérifier la communication entre les machines.

OBJECTIFS DU TD

Les principaux objectifs de ce travail sont :

Savoir attribuer des adresses IP valides aux machines.

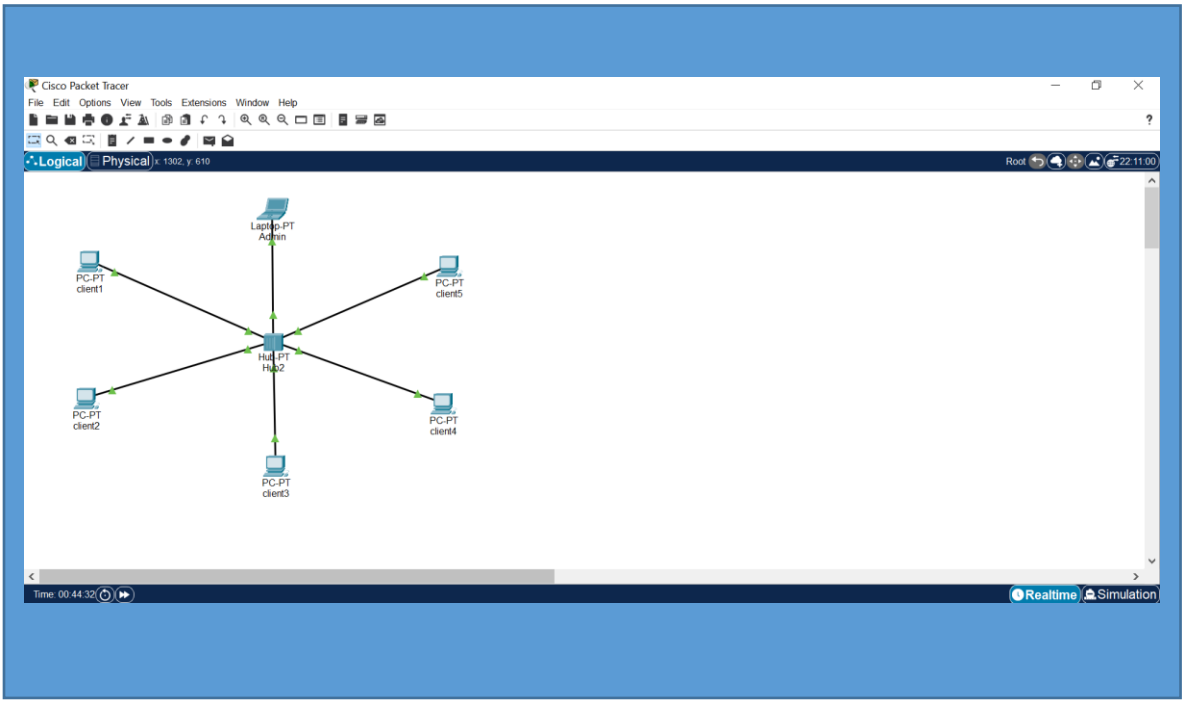
Comprendre l'adressage IPv4 et IPv6.

Configurer des adresses IP sur les hôtes.

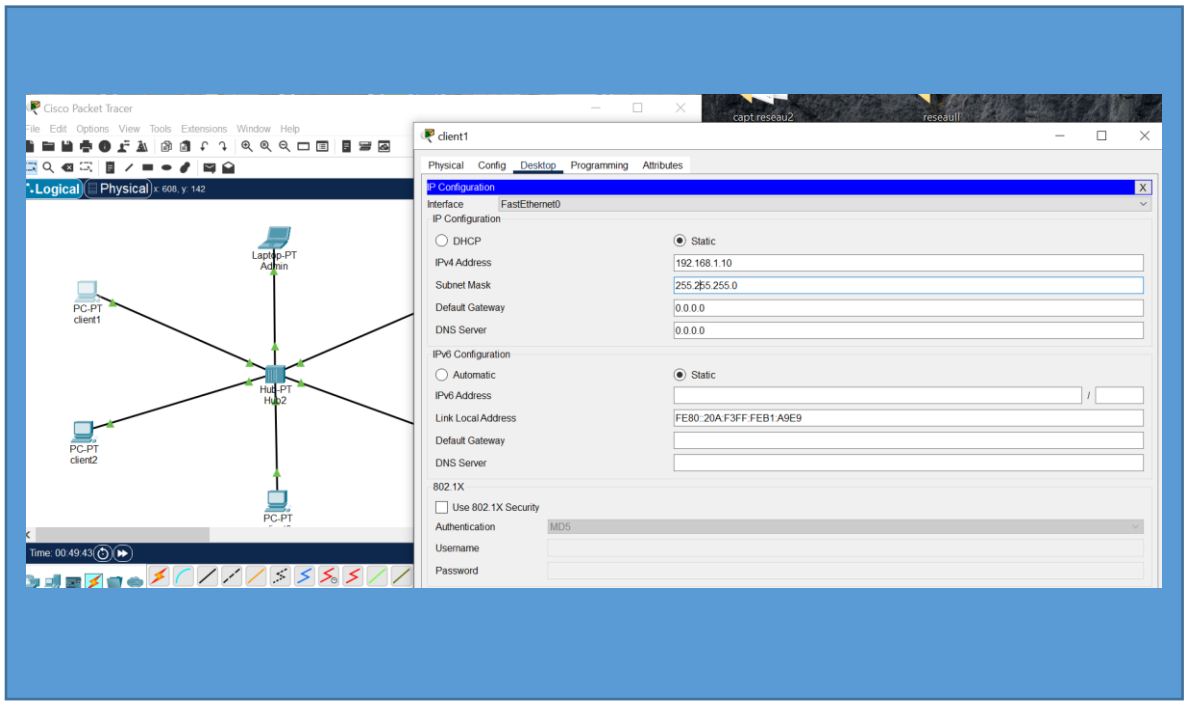
Reproduire la première topologie dans Cisco Packet Tracer, attribuer les adresses IPv4 indiquées dans le tableau et vérifier la connectivité entre les équipements.

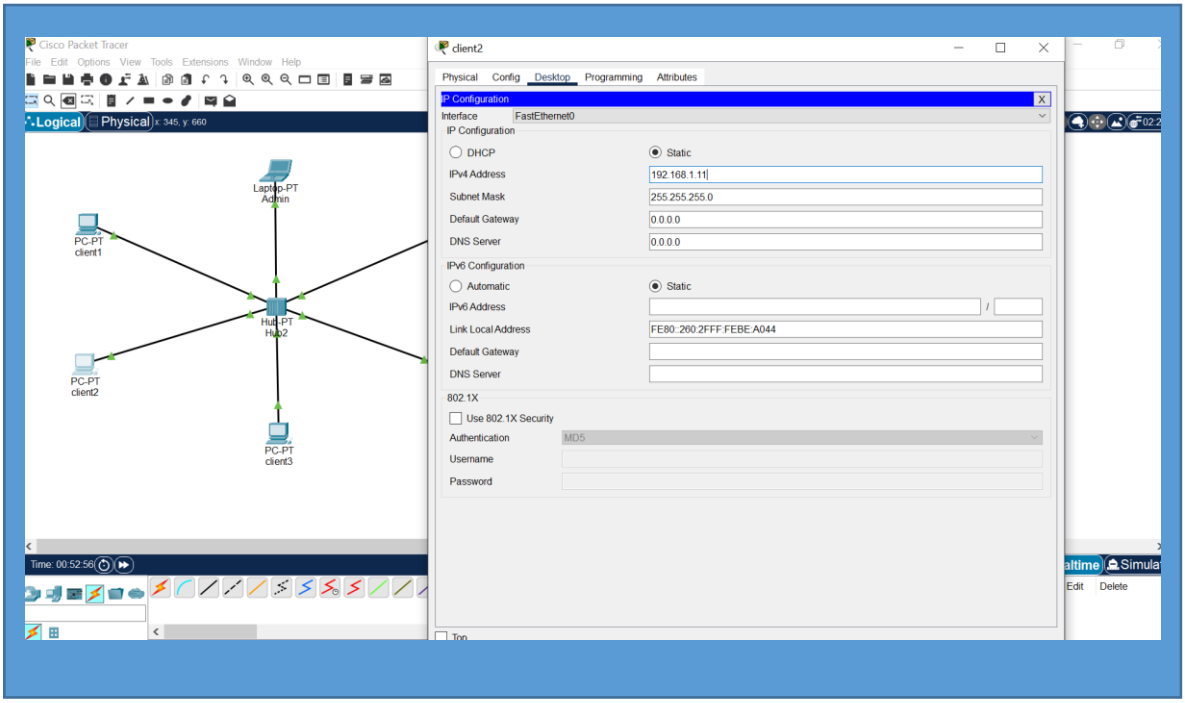
Création de la topologie dans Cisco Packet Tracer.

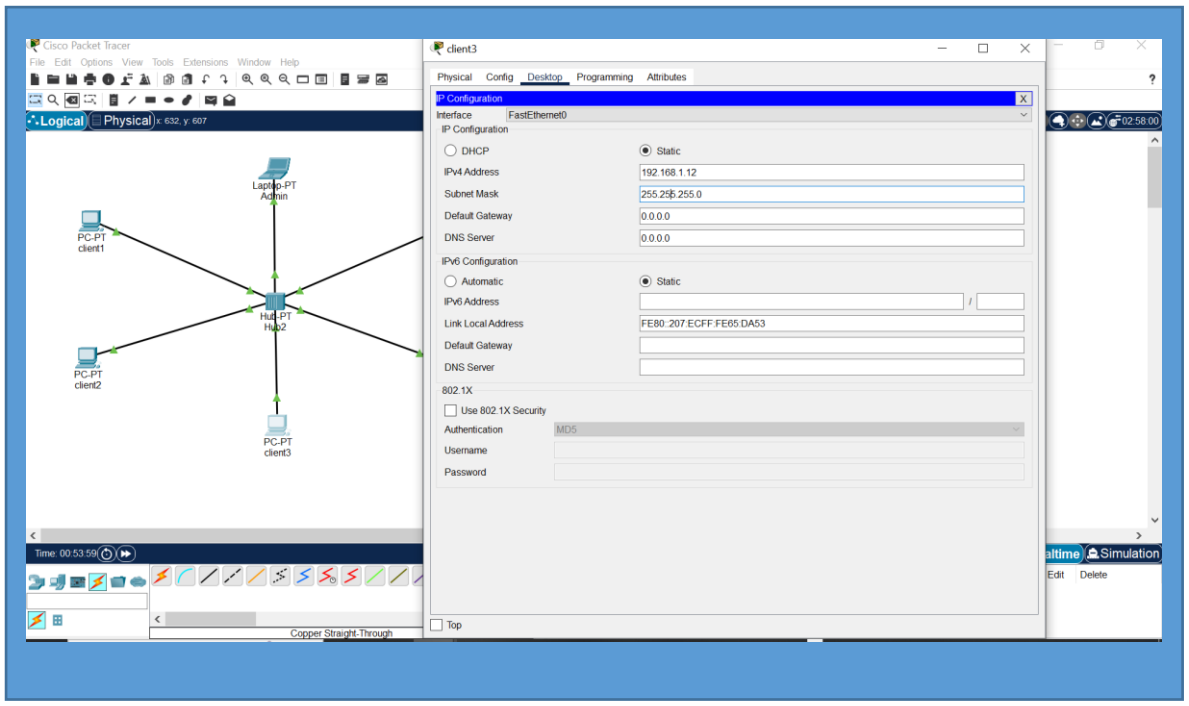
Connexion des équipements à l'aide des câbles appropriés

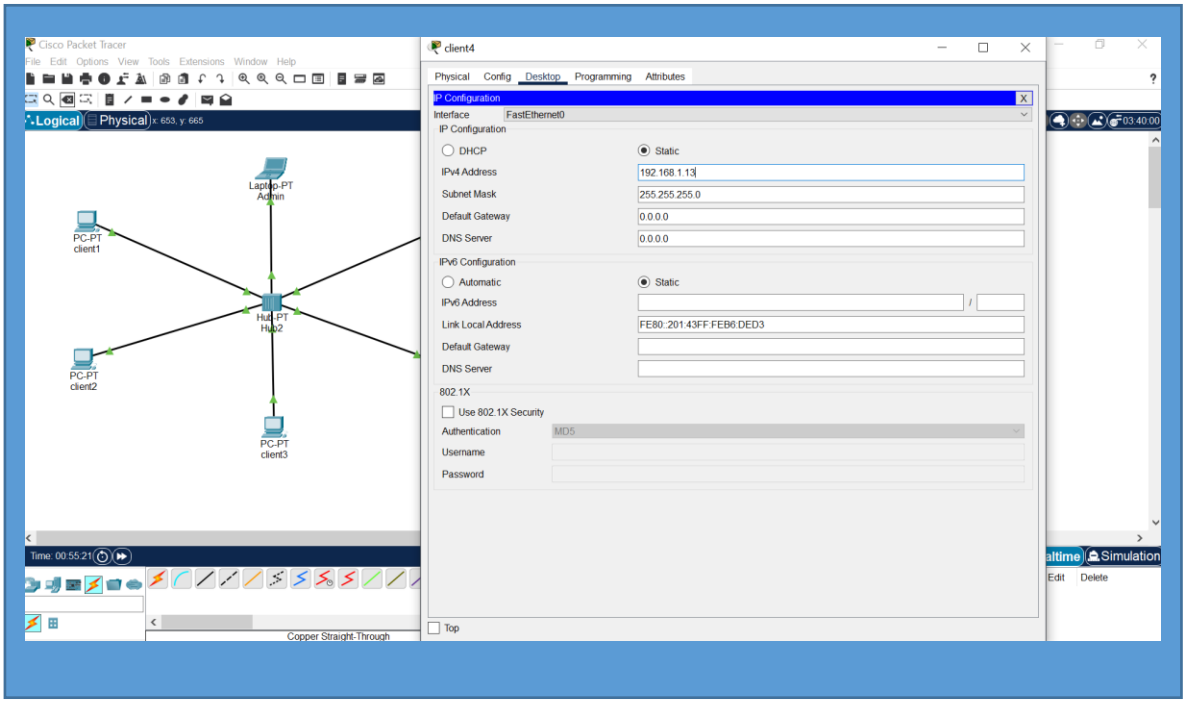


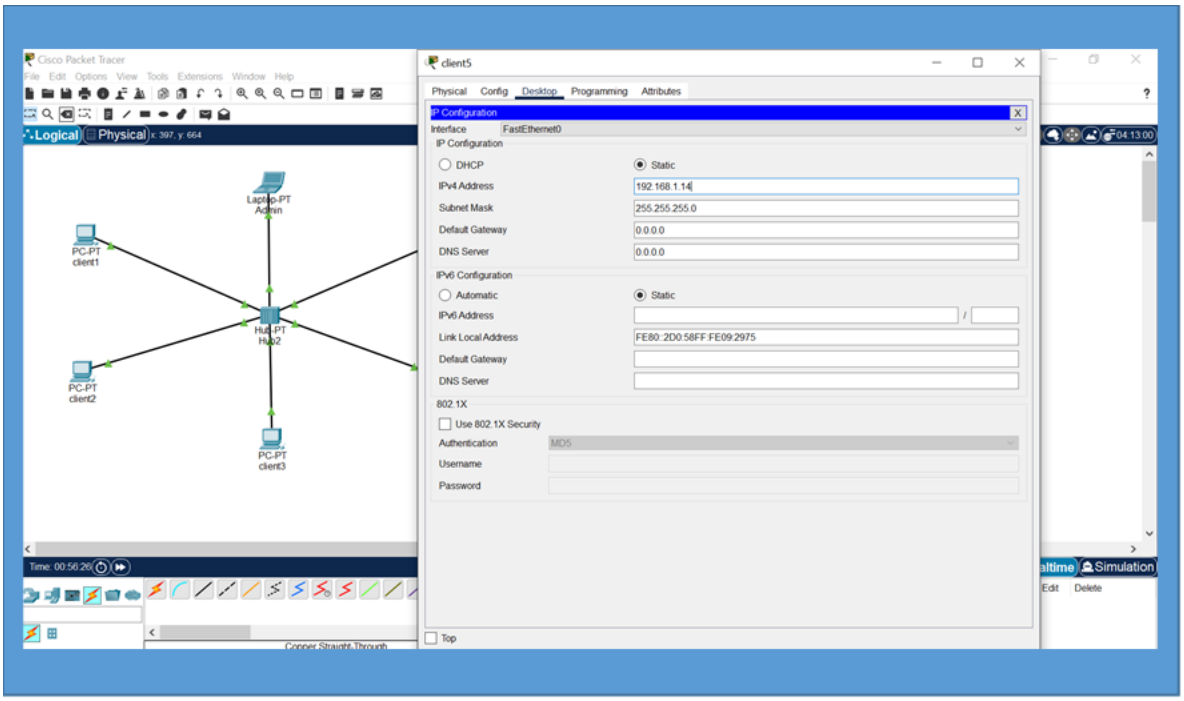
2. Configuration des adresses IPv4.

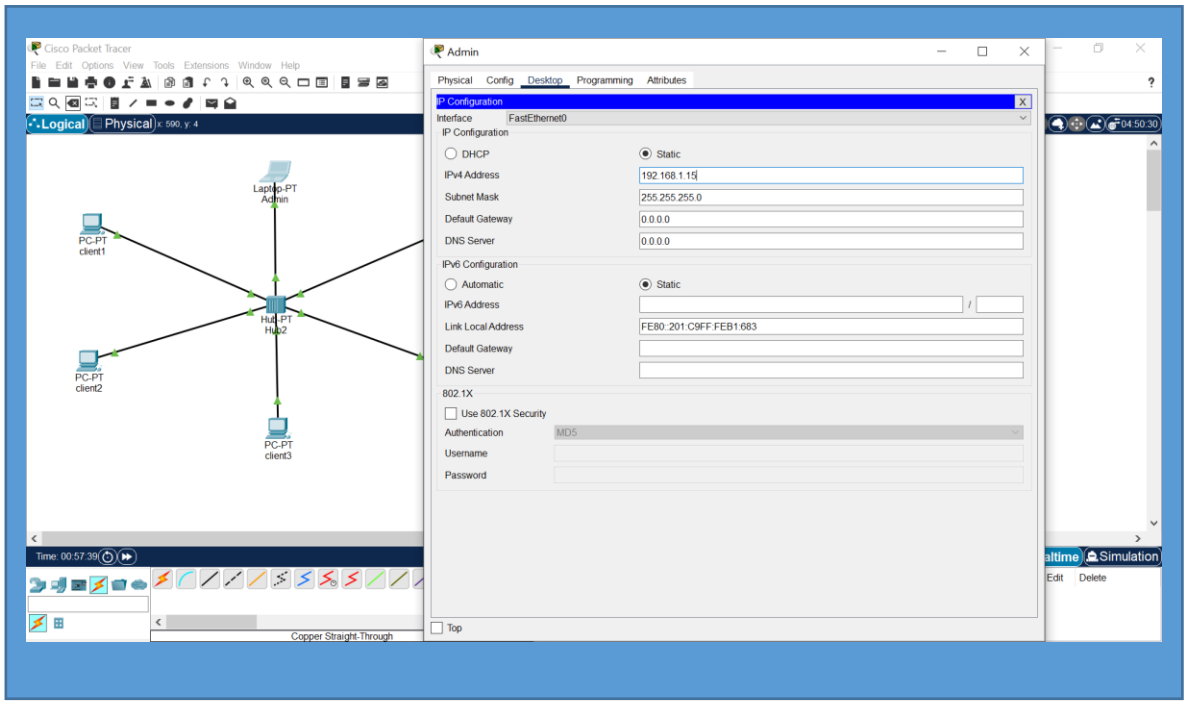




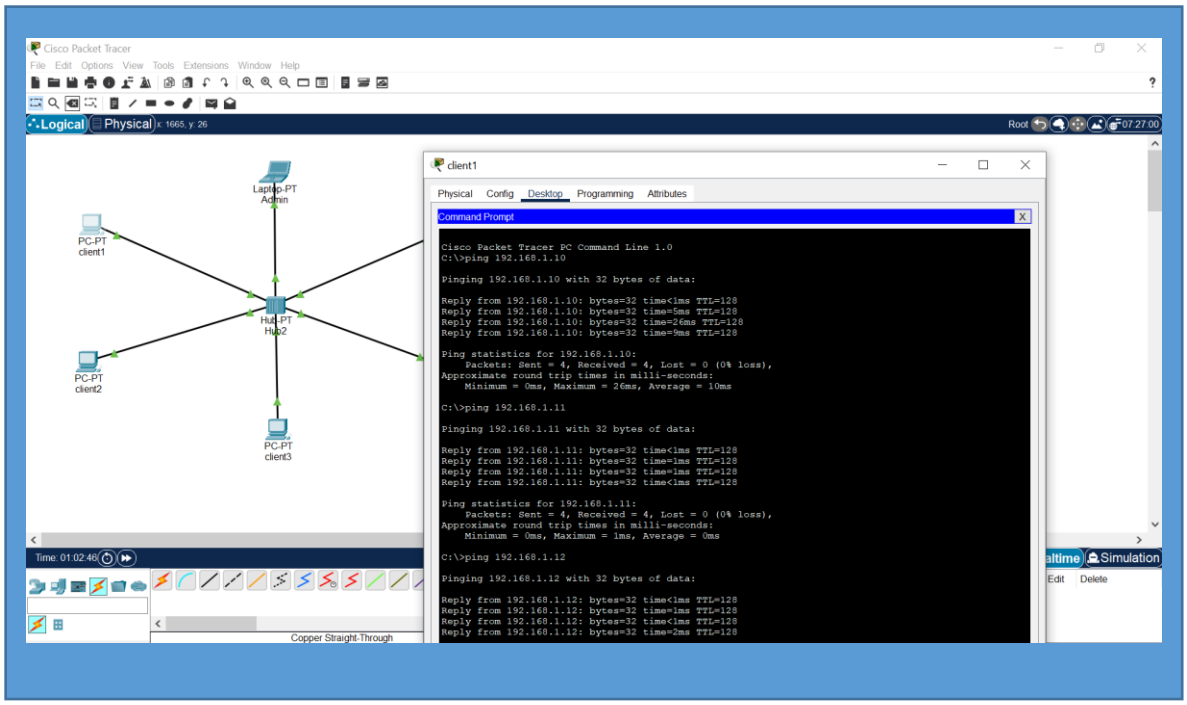

Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical Physical x: 632, y: 607
Laptop-PT Admin
PC-PT client1
Hub-PT Hub2
PC-PT client2
PC-PT client3
Time: 00:53:59
Copper Straight-Through
client3
Physical Config Desktop Programming Attributes
IP Configuration
Interface FastEthernet0
DHCP
Static
IPv4 Address 192.168.1.12
Subnet Mask 255.255.255.0
Default Gateway 0.0.0.0
DNS Server 0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address FE80::207:ECFF:FE65:DA53
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication MD5
Username
Password
Top
02:58:00
Simulation
Edit Delete

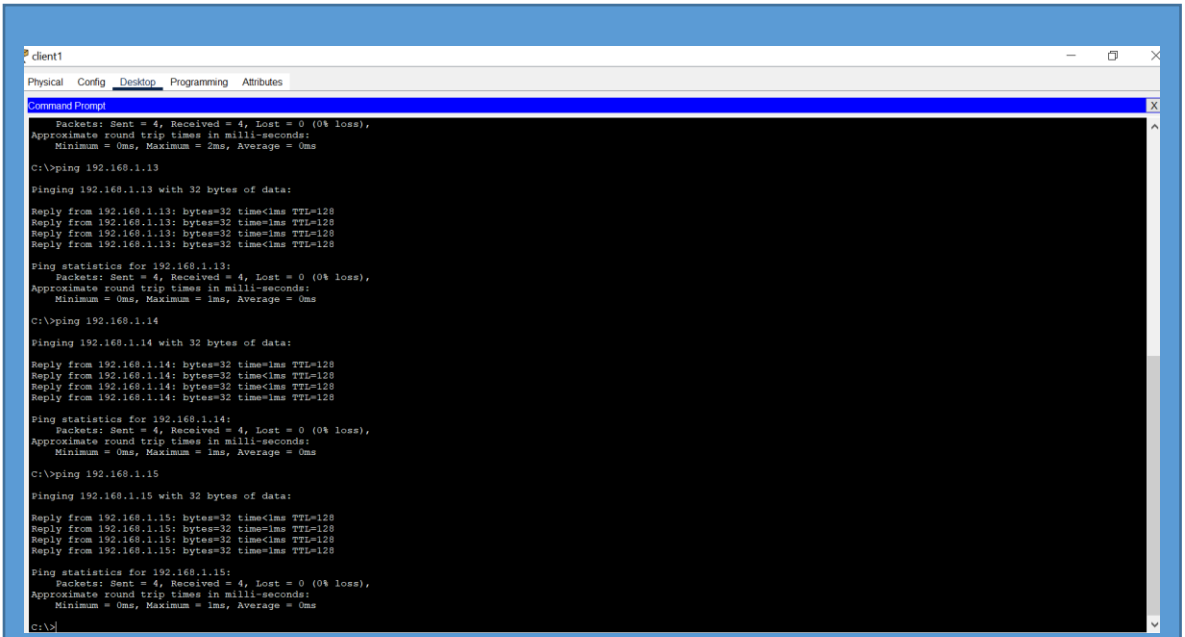

Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical Physical x: 653, y: 665
Laptop-PT Admin
PC-PT client1
Hub-PT Hub2
PC-PT client2
PC-PT client3
Time: 00:55:21
Copper Straight-Through
client4
Physical Config Desktop Programming Attributes
IP Configuration
Interface FastEthernet0
DHCP
Static
IPv4 Address 192.168.1.13
Subnet Mask 255.255.255.0
Default Gateway 0.0.0.0
DNS Server 0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address FE80::201:43FF:FEB6:DED3
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication MD5
Username
Password
Top
03:40:00
Simulation
Edit Delete


Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical Physical x: 397, y: 664
Laptop-PT Admin
PC-PT client1
Hub-PT Hub2
PC-PT client2
PC-PT client3
Time: 00:56:26
Copper Straight-Through
client5
Physical Config Desktop Programming Attributes
IP Configuration
Interface FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address 192.168.1.14
Subnet Mask 255.255.255.0
Default Gateway 0.0.0.0
DNS Server 0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address FE80::2D0:58FF:FE09:2975
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication MD5
Username
Password
Top
04:13:00
Simulation
Edit Delete


Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical Physical x: 590, y: 4
Laptop-PT Admin
PC-PT client1
Hub-PT Hub2
PC-PT client2
PC-PT client3
Time: 00:57:39
Copper Straight-Through
Admin
Physical Config Desktop Programming Attributes
IP Configuration
Interface FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address 192.168.1.15
Subnet Mask 255.255.255.0
Default Gateway 0.0.0.0
DNS Server 0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address FE80::201:C9FF:FEB1:683
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication MD5
Username
Password
Top
04:50:30
Simulation
Edit Delete

3 -Vérification de la connectivité avec la commande Ping.

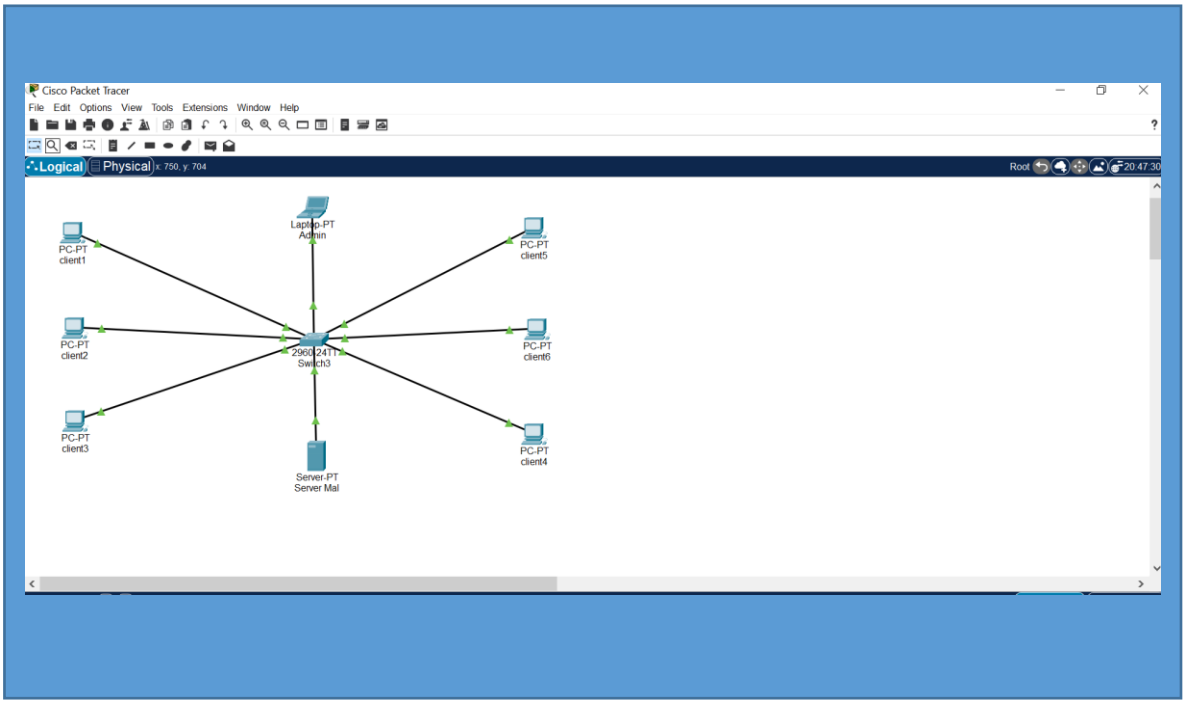




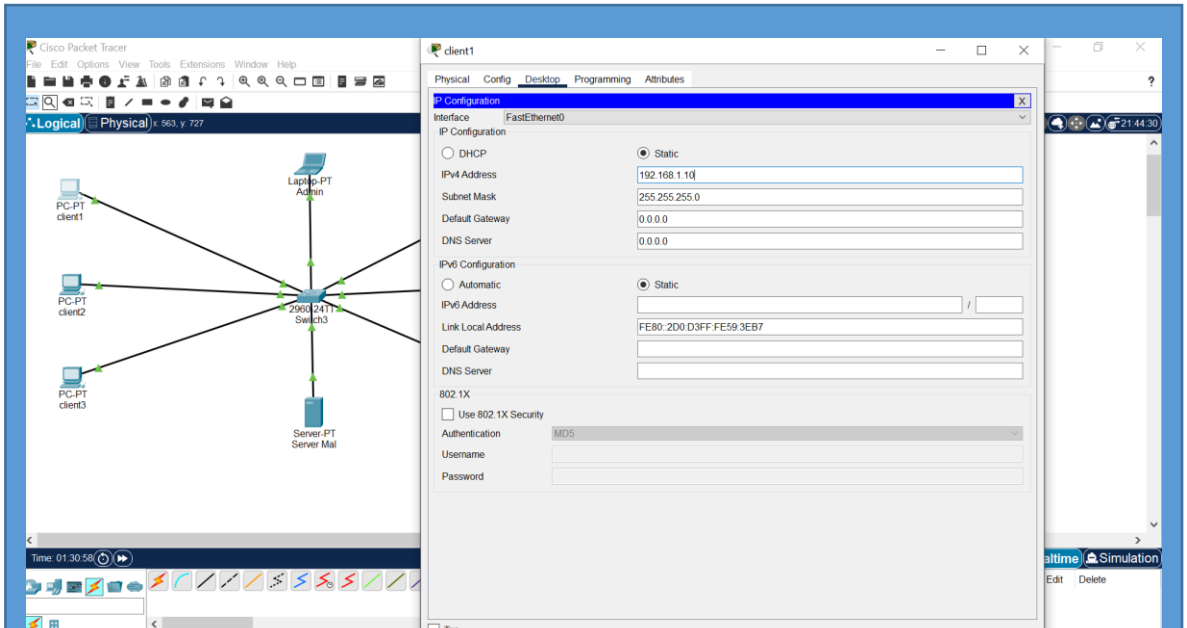
NTÂCHE 2 : CONFIGURATION IPv4

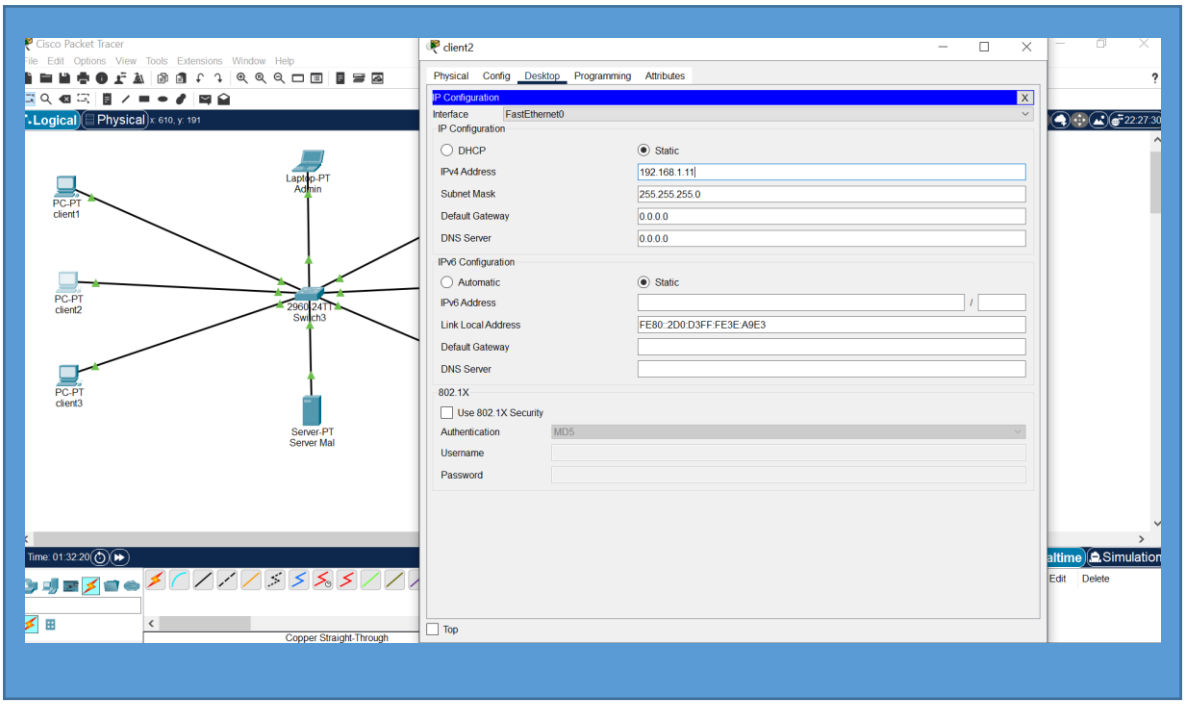
Reproduire la deuxième topologie et configurer les adresses IPv4 conformément au tableau fourni.

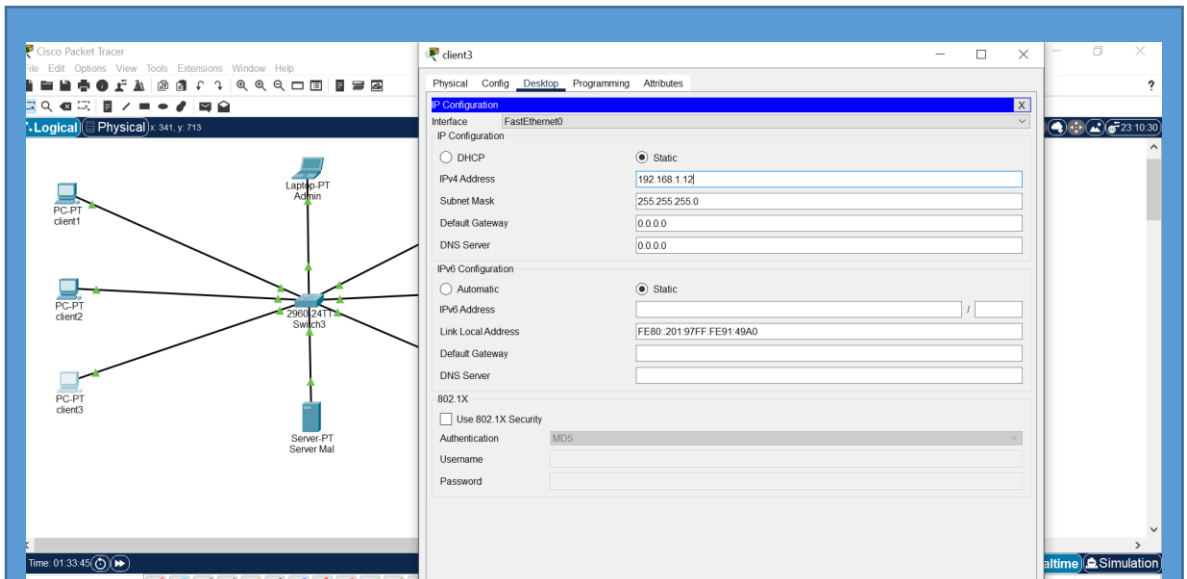
Reproduiction de la topologie et connection des differents equipements.

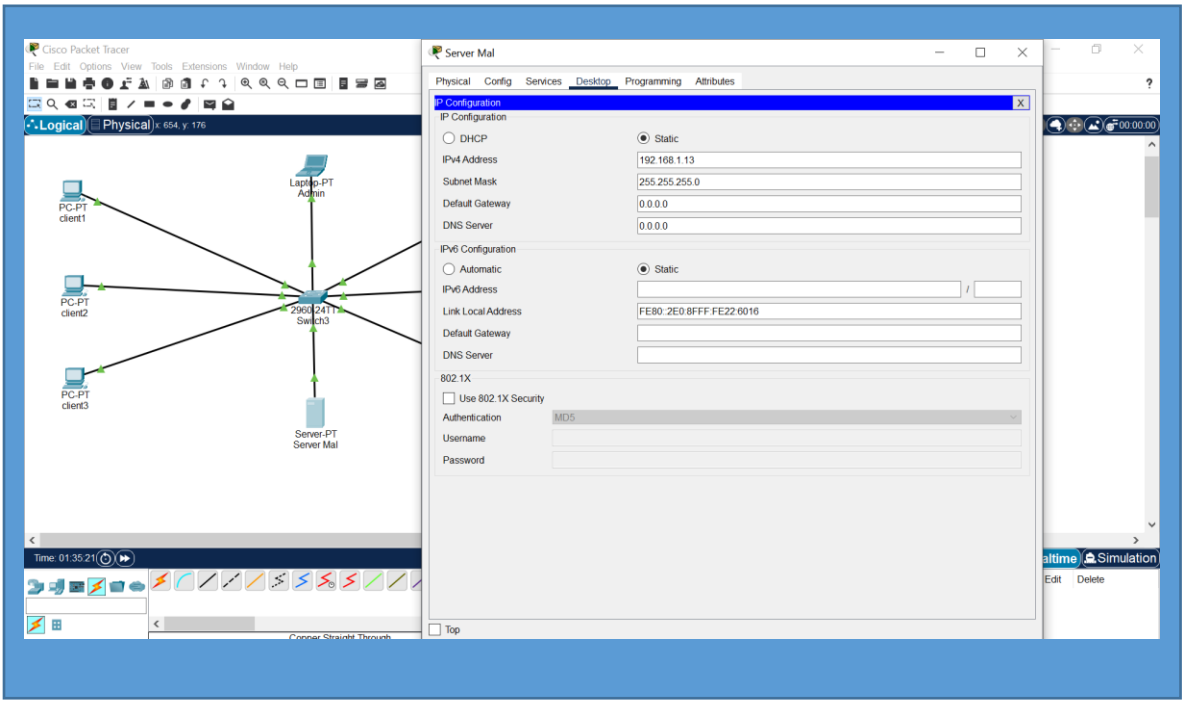


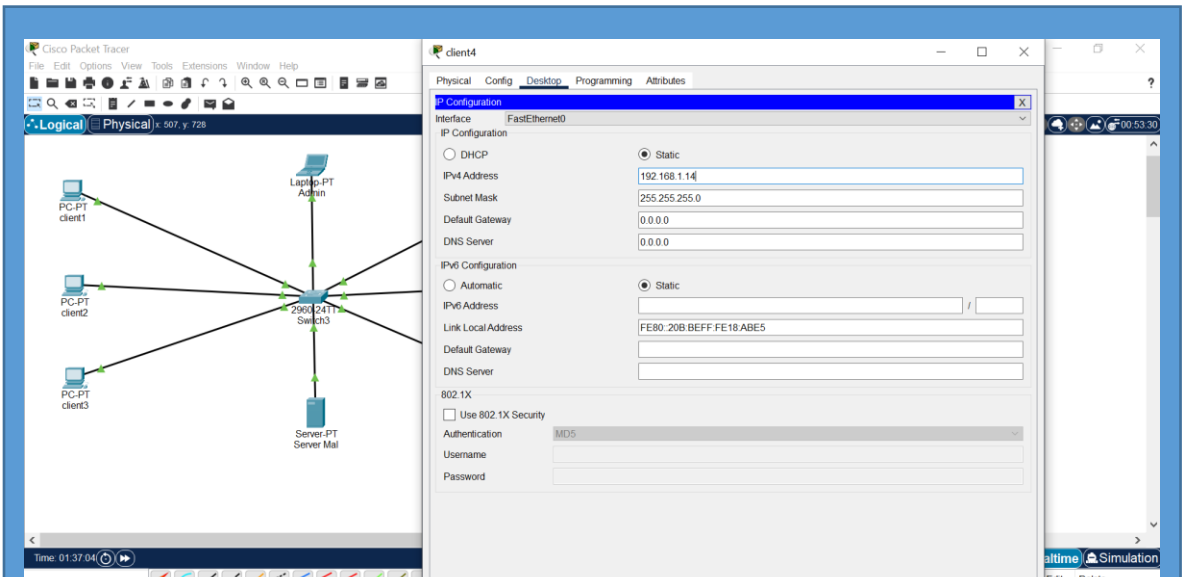
Attribution des adressees IPV4

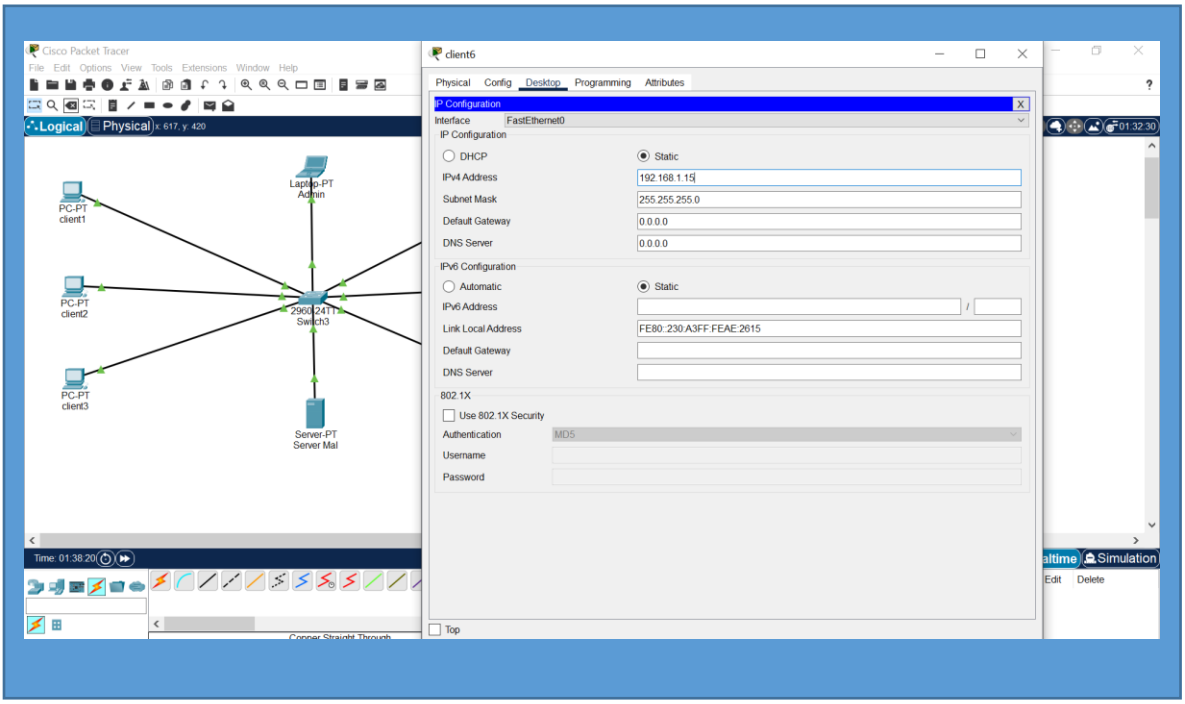


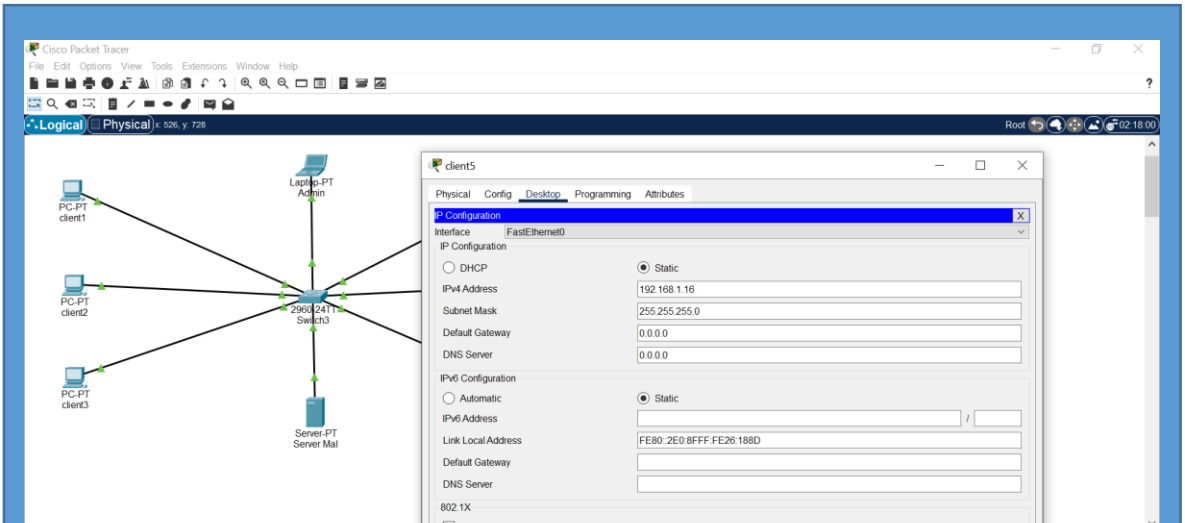

Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical Physical x: 610, y: 191
Laptop-PT Admin
PC-PT client1
PC-PT client2
PC-PT client3
2960-24TT Switch3
Server-PT Server Mal
Time: 01:32:20
Copper Straight-Through
client2
Physical Config Desktop Programming Attributes
IP Configuration
Interface FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address 192.168.1.11
Subnet Mask 255.255.255.0
Default Gateway 0.0.0.0
DNS Server 0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address FE80::2D0:D3FF:FE3E:A9E3
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication MD5
Username
Password
Top
Edit Delete
Simulation

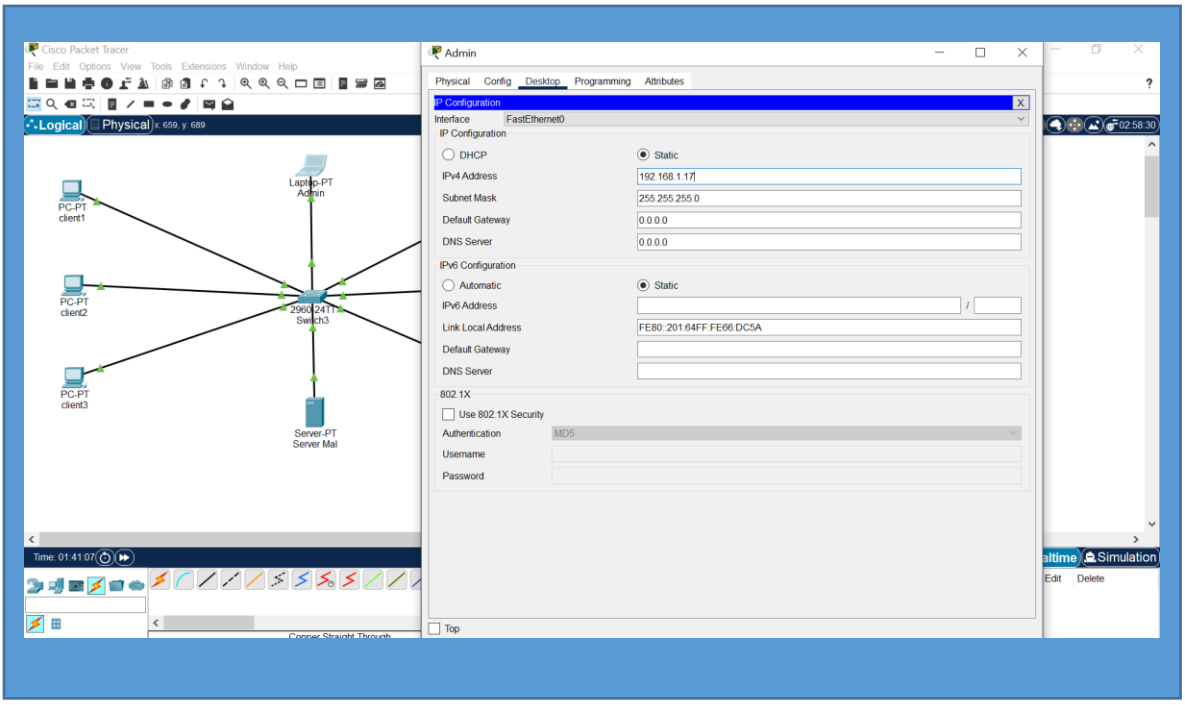

Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical Physical x: 341, y: 713
Laptop-PT Admin
PC-PT client1
PC-PT client2
PC-PT client3
2960-24TT Switch3
Server-PT Server Mal
Time: 01:33:45
client3
Physical Config Desktop Programming Attributes
IP Configuration
Interface FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address 192.168.1.12
Subnet Mask 255.255.255.0
Default Gateway 0.0.0.0
DNS Server 0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address FE80::201:97FF:FE91:49A0
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication MD5
Username
Password
Simulation


Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical Physical x: 654, y: 176
LapMo-PT
Admin
PC-PT
client1
PC-PT
client2
PC-PT
client3
2960-24TT
Switch3
Server-PT
Server Mal
Time: 01:35:21
Server Mal
Physical Config Services Desktop Programming Attributes
IP Configuration
IP Configuration
DHCP
Static
IPv4 Address 192.168.1.13
Subnet Mask 255.255.255.0
Default Gateway 0.0.0.0
DNS Server 0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address FE80::2E0:8FFF:FE22:6016
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication MD5
Username
Password
Top
00:00:00
Simulation
Edit Delete


Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical Physical x: 507, y: 728
LapMo-PT
Admin
PC-PT
client1
PC-PT
client2
PC-PT
client3
2960-24TT
Switch3
Server-PT
Server Mal
Time: 01:37:04
client4
Physical Config Desktop Programming Attributes
IP Configuration
Interface FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address 192.168.1.14
Subnet Mask 255.255.255.0
Default Gateway 0.0.0.0
DNS Server 0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address FE80::20B:BEFF:FE18:ABE5
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication MD5
Username
Password
00:53:30
Simulation

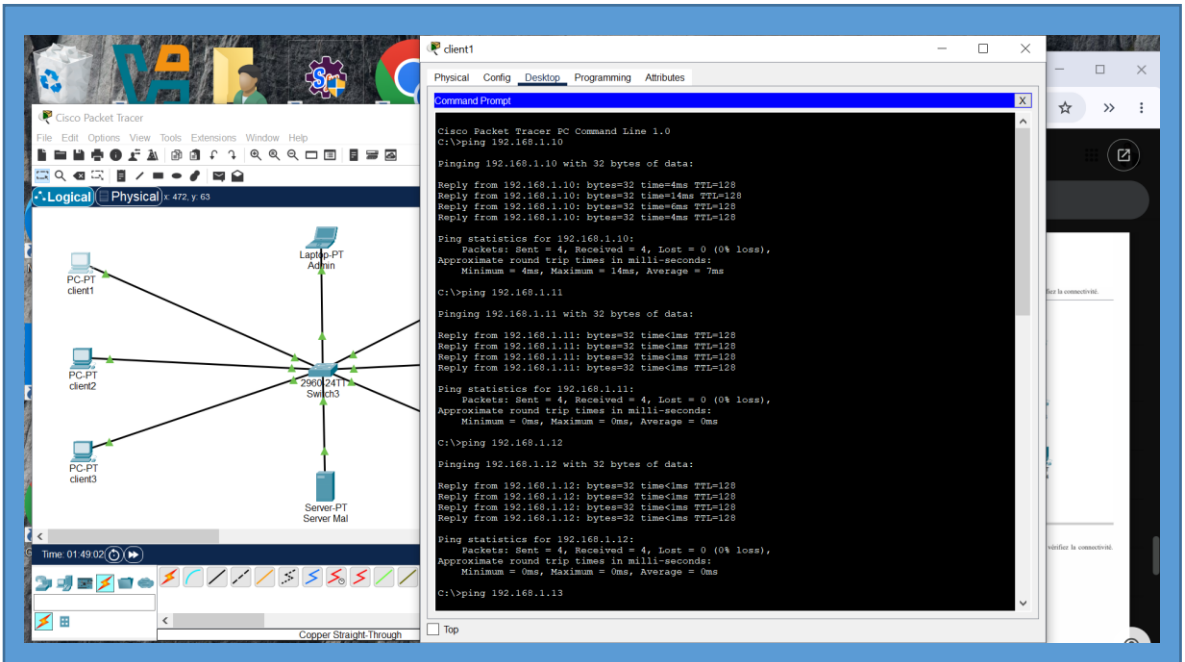

Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical Physical x: 617, y: 420
Laptop-PT Admin
PC-PT client1
PC-PT client2
PC-PT client3
2960-24TT Switch3
Server-PT Server Mal
Time: 01:38:20
client6
Physical Config Desktop Programming Attributes
IP Configuration
Interface FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address 192.168.1.15
Subnet Mask 255.255.255.0
Default Gateway 0.0.0.0
DNS Server 0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address FE80::230:A3FF:FEAE:2615
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication MD5
Username
Password
Top
01:32:30
Simulation
Edit Delete

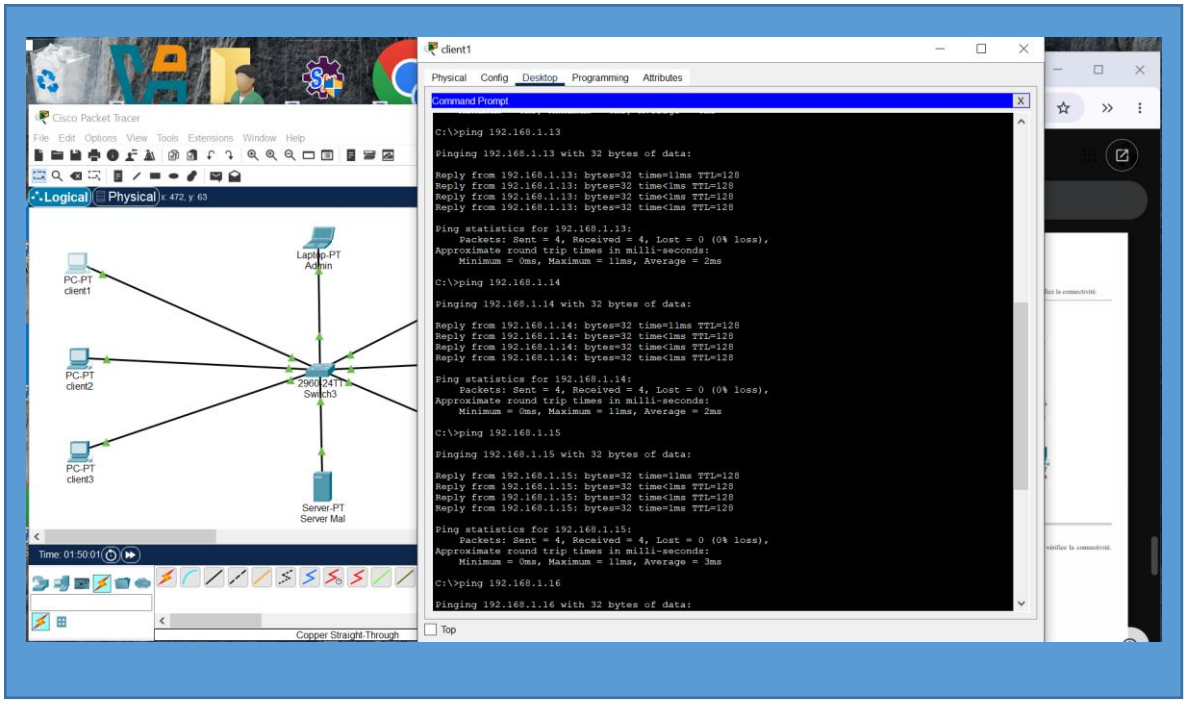

Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical Physical x: 526, y: 728
Root
02:18:00
Laptop-PT Admin
PC-PT client1
PC-PT client2
PC-PT client3
2960-24TT Switch3
Server-PT Server Mal
client5
Physical Config Desktop Programming Attributes
IP Configuration
Interface FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address 192.168.1.16
Subnet Mask 255.255.255.0
Default Gateway 0.0.0.0
DNS Server 0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address FE80::2E0:8FFF:FE26:188D
Default Gateway
DNS Server
802.1X

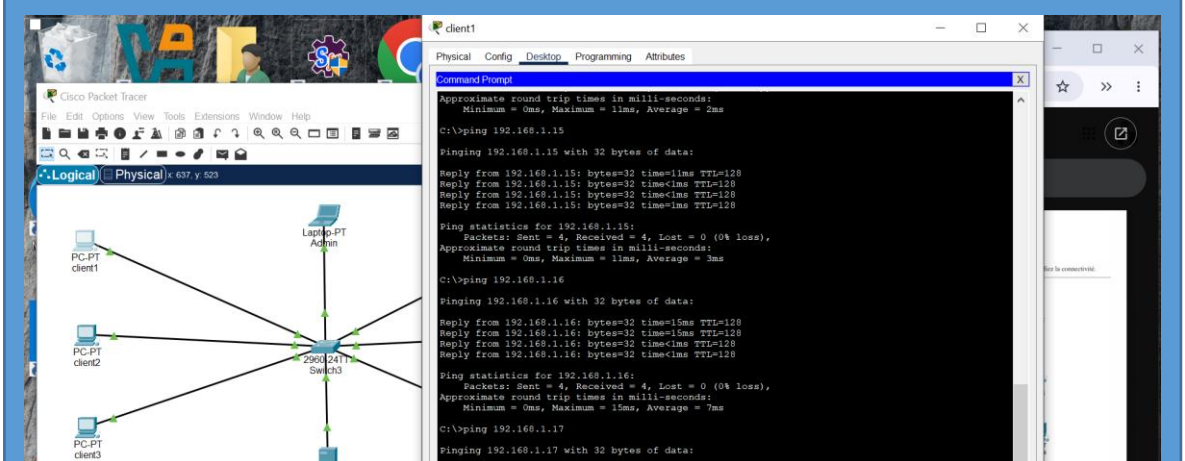


3 -Vérification de la connectivité avec la commande Ping

1.



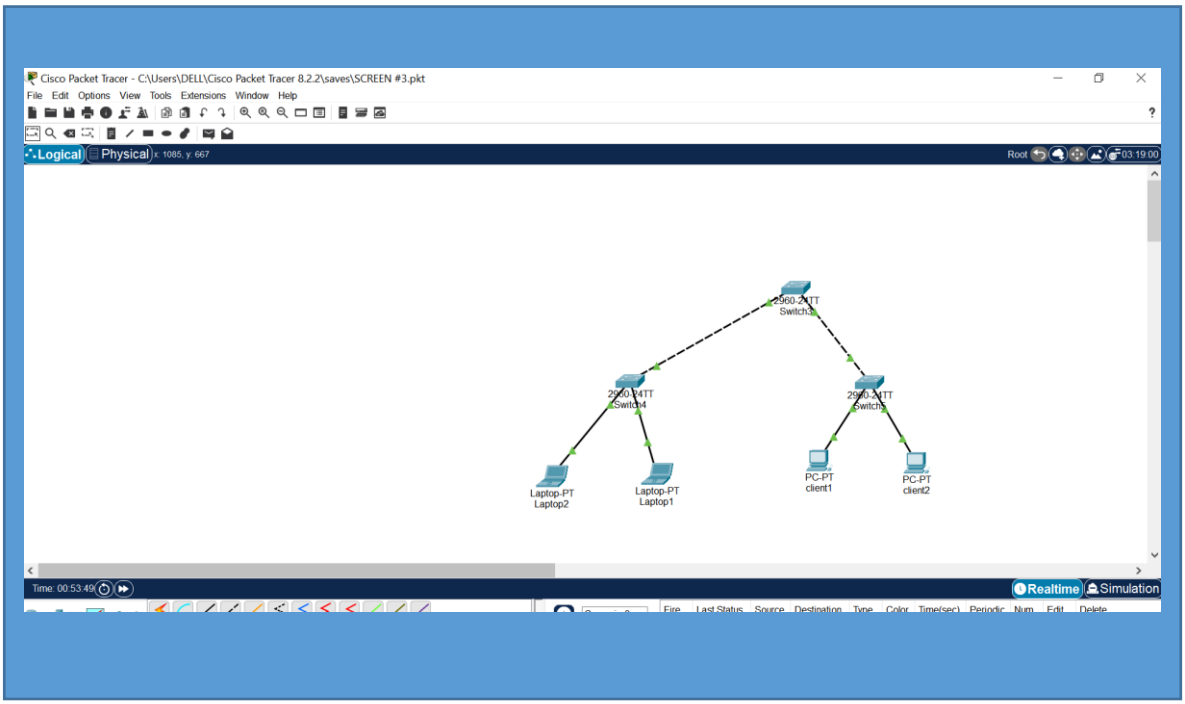

client1
Physical
Config
Desktop
Programming
Attributes
Command Prompt
C:\>ping 192.168.1.13
Pinging 192.168.1.13 with 32 bytes of data:
Reply from 192.168.1.13: bytes=32 time=11ms TTL=128
Reply from 192.168.1.13: bytes=32 time<1ms TTL=128
Reply from 192.168.1.13: bytes=32 time<1ms TTL=128
Reply from 192.168.1.13: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.1.13:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 11ms, Average = 2ms
C:\>ping 192.168.1.14
Pinging 192.168.1.14 with 32 bytes of data:
Reply from 192.168.1.14: bytes=32 time=11ms TTL=128
Reply from 192.168.1.14: bytes=32 time<1ms TTL=128
Reply from 192.168.1.14: bytes=32 time<1ms TTL=128
Reply from 192.168.1.14: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.1.14:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 11ms, Average = 2ms
C:\>ping 192.168.1.15
Pinging 192.168.1.15 with 32 bytes of data:
Reply from 192.168.1.15: bytes=32 time=11ms TTL=128
Reply from 192.168.1.15: bytes=32 time<1ms TTL=128
Reply from 192.168.1.15: bytes=32 time<1ms TTL=128
Reply from 192.168.1.15: bytes=32 time=1ms TTL=128
Ping statistics for 192.168.1.15:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 11ms, Average = 3ms
C:\>ping 192.168.1.16
Pinging 192.168.1.16 with 32 bytes of data:
Top
Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical
Physical
Laptop-PT
Admin
PC-PT
client1
PC-PT
client2
PC-PT
client3
2960-24TT
Switch3
Server-PT
Server Mal
Time: 01:50:01
Copper Straight-Through


client1
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 11ms, Average = 2ms
C:\>ping 192.168.1.15
Pinging 192.168.1.15 with 32 bytes of data:
Reply from 192.168.1.15: bytes=32 time=11ms TTL=128
Reply from 192.168.1.15: bytes=32 time<1ms TTL=128
Reply from 192.168.1.15: bytes=32 time<1ms TTL=128
Reply from 192.168.1.15: bytes=32 time=1ms TTL=128
Ping statistics for 192.168.1.15:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 11ms, Average = 3ms
C:\>ping 192.168.1.16
Pinging 192.168.1.16 with 32 bytes of data:
Reply from 192.168.1.16: bytes=32 time=15ms TTL=128
Reply from 192.168.1.16: bytes=32 time=15ms TTL=128
Reply from 192.168.1.16: bytes=32 time<1ms TTL=128
Reply from 192.168.1.16: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.1.16:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 15ms, Average = 7ms
C:\>ping 192.168.1.17
Pinging 192.168.1.17 with 32 bytes of data:
Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical
Physical
Laptop-PT
Admin
PC-PT
client1
PC-PT
client2
PC-PT
client3
2960-24TT
Switch3

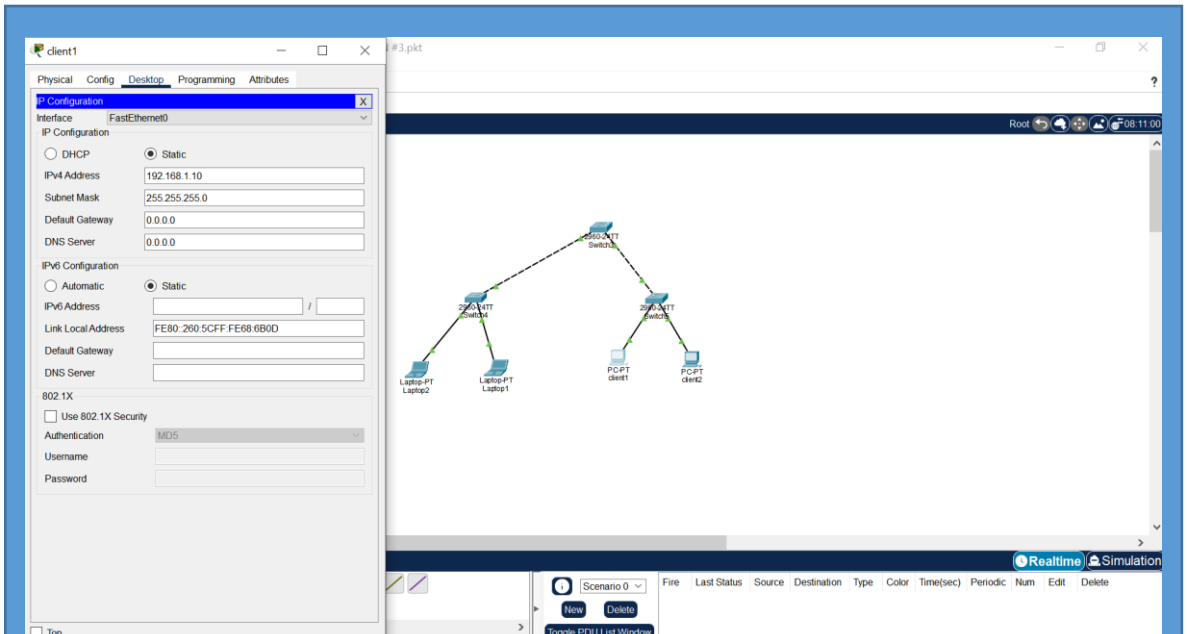
TÂCHE 2 : CONFIGURATION IPv4

Reproduire la deuxième topologie et configurer les adresses IPv4 conformément au tableau fourni.

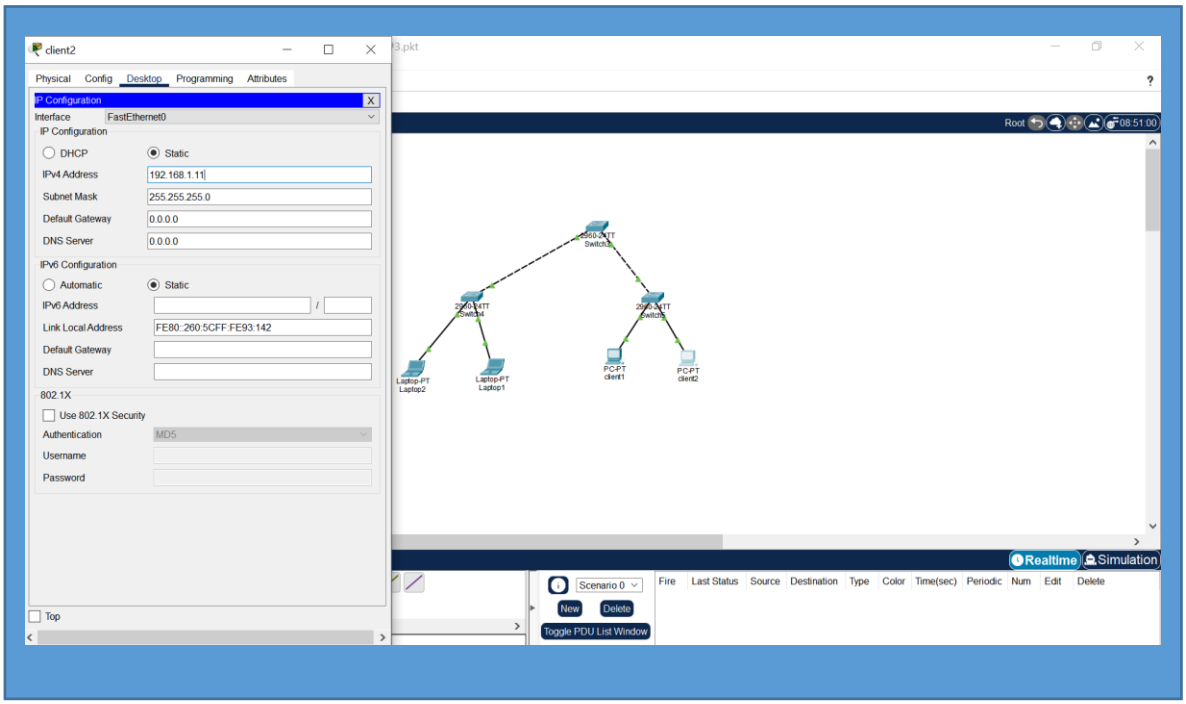
Choix d'une topologie

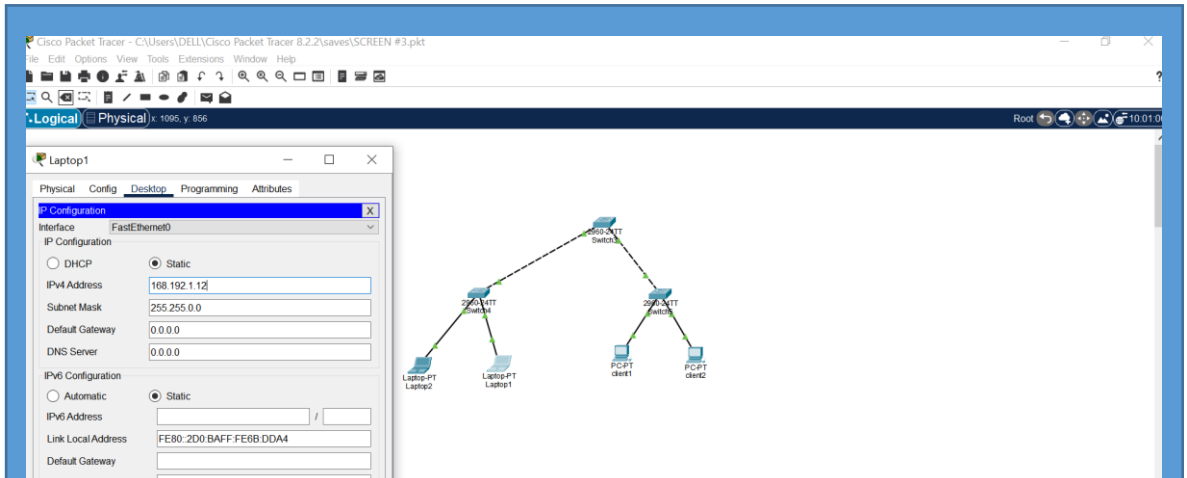


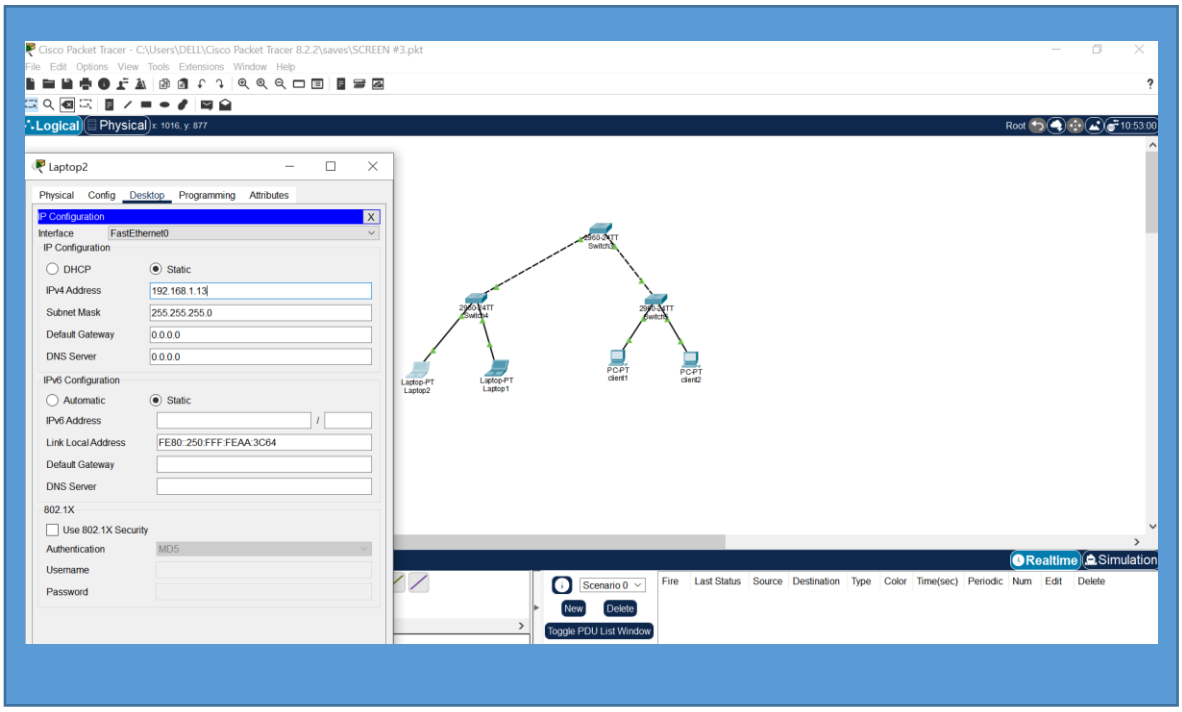
Configuration des addresses IPV4



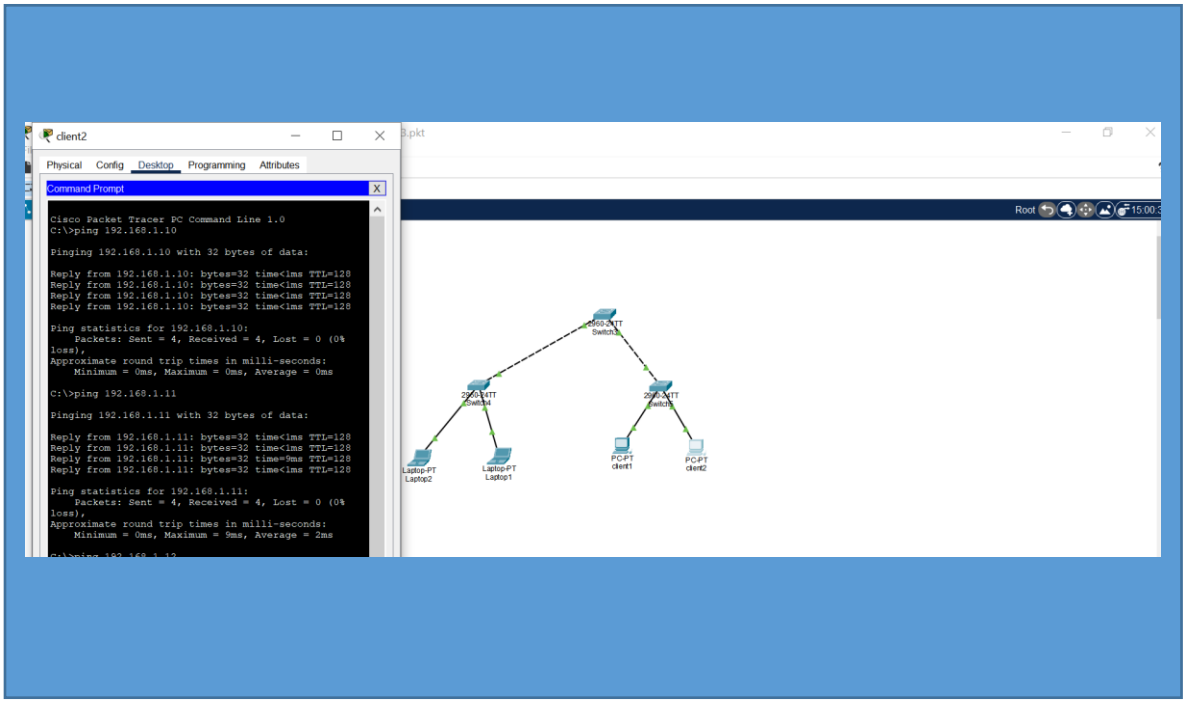
2. N

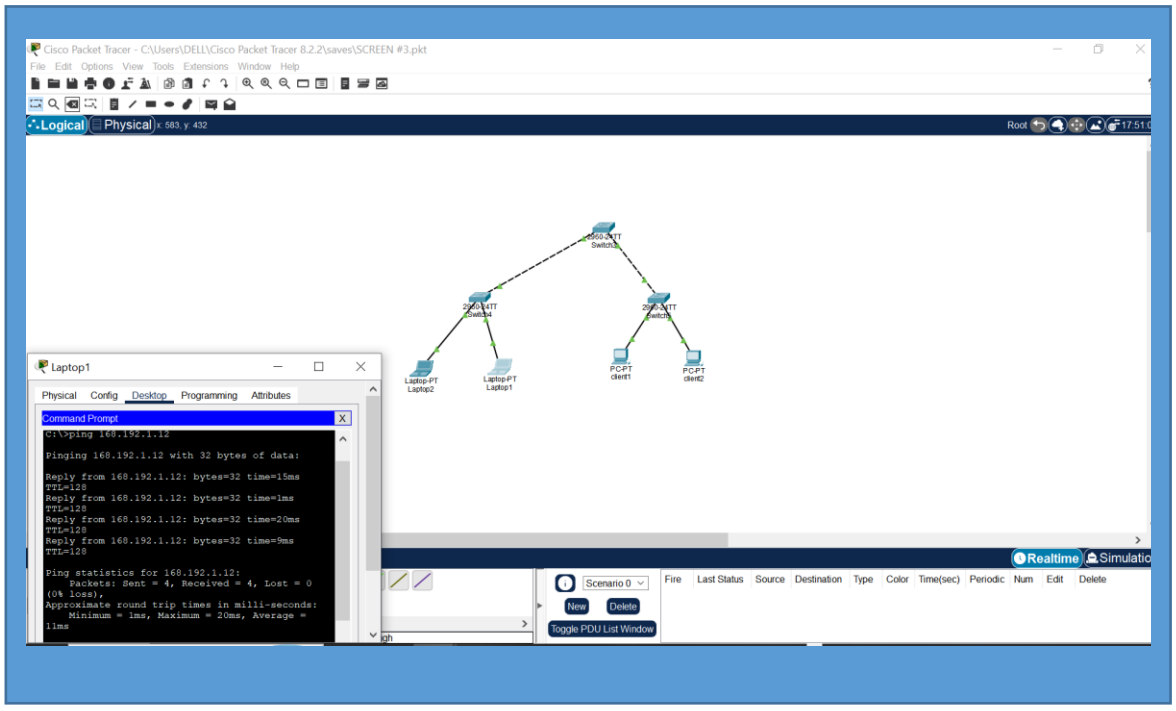

client2
Physical Config Desktop Programming Attributes
IP Configuration
Interface FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address 192.168.1.11
Subnet Mask 255.255.255.0
Default Gateway 0.0.0.0
DNS Server 0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address FE80::260:5CFF:FE93:142
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication MD5
Username
Password
Top
Root
Realtime
Simulation
Scenario 0
New
Delete
Toggle PDU List Window
Fire Last Status Source Destination Type Color Time(sec) Periodic Num Edit Delete

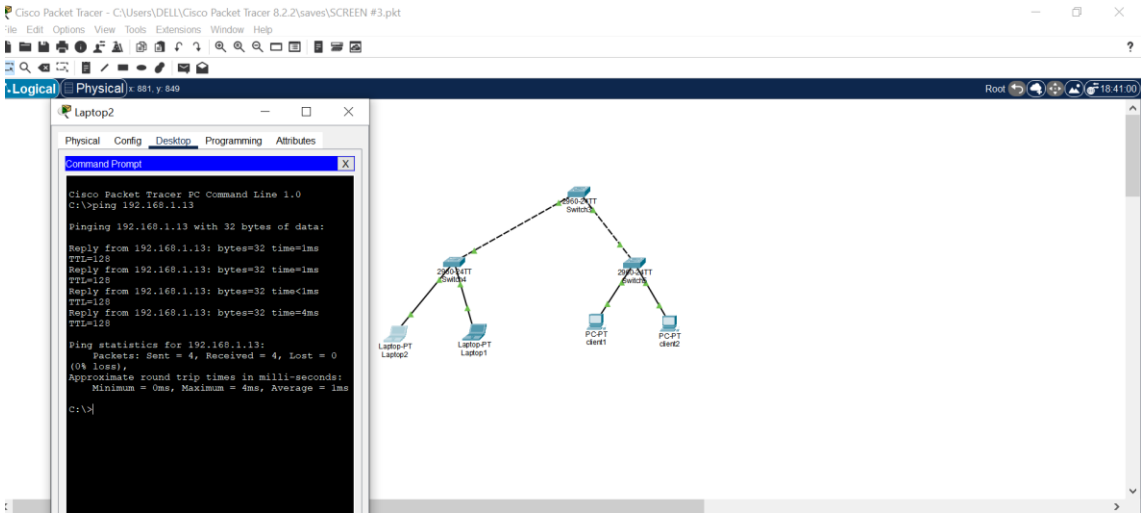

Logical
Physical
Root
Laptop1
Physical Config Desktop Programming Attributes
IP Configuration
Interface FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address 168.192.1.12
Subnet Mask 255.255.0.0
Default Gateway 0.0.0.0
DNS Server 0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address FE80::2D0:BAFF:FE6B:DDA4
Default Gateway



Relisation des tests Ping



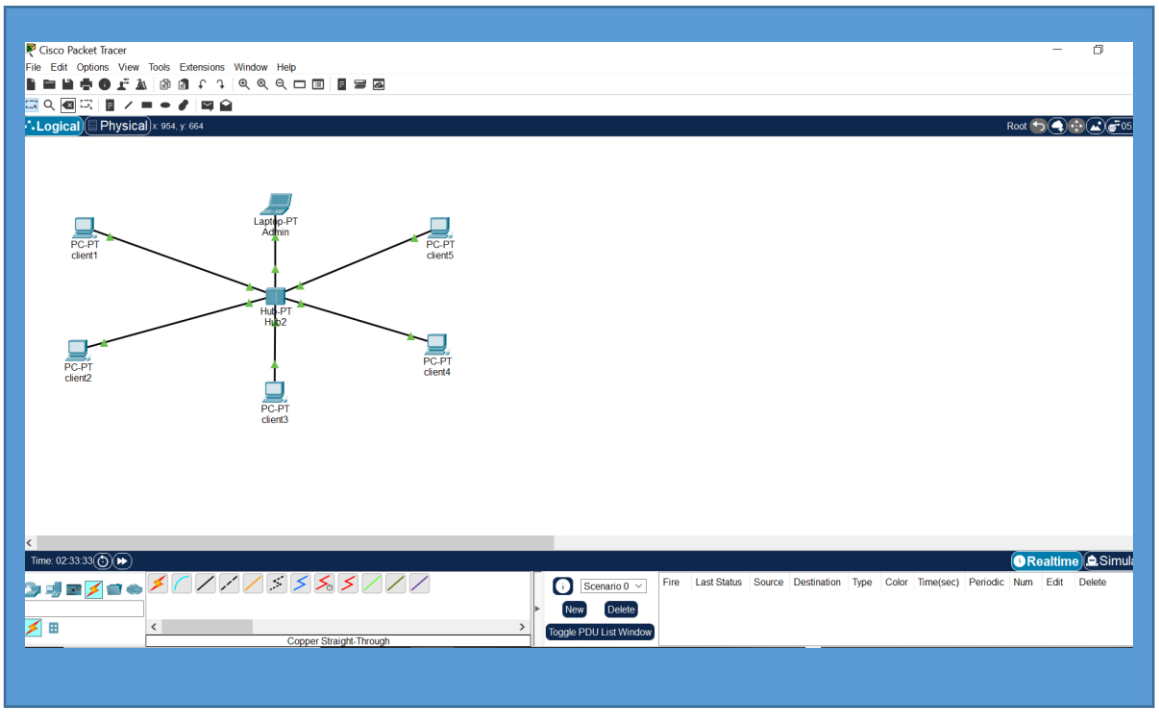

Cisco Packet Tracer - C:\Users\DELL\Cisco Packet Tracer 8.2.2\saves\SCREEN #3.pkt
File Edit Options View Tools Extensions Window Help
Logical Physical x: 583, y: 432
Root
Laptop1
Physical Config Desktop Programming Attributes
Command Prompt
C:\>ping 168.192.1.12
Pinging 168.192.1.12 with 32 bytes of data:
Reply from 168.192.1.12: bytes=32 time=15ms TTL=128
Reply from 168.192.1.12: bytes=32 time=1ms TTL=128
Reply from 168.192.1.12: bytes=32 time=20ms TTL=128
Reply from 168.192.1.12: bytes=32 time=9ms TTL=128
Ping statistics for 168.192.1.12:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 20ms, Average = 11ms
Laptop-PT Laptop2
Laptop-PT Laptop1
PC-PT client1
PC-PT client2
Realtime
Scenario 0
New
Delete
Toggle PDU List Window
Fire Last Status Source Destination Type Color Time(sec) Periodic Num Edit Delete


Cisco Packet Tracer - C:\Users\DELL\Cisco Packet Tracer 8.2.2\saves\SCREEN #3.pkt
Logical Physical x: 881, y: 849
Root
Laptop2
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.13
Pinging 192.168.1.13 with 32 bytes of data:
Reply from 192.168.1.13: bytes=32 time=1ms TTL=128
Reply from 192.168.1.13: bytes=32 time=1ms TTL=128
Reply from 192.168.1.13: bytes=32 time<1ms TTL=128
Reply from 192.168.1.13: bytes=32 time=4ms TTL=128
Ping statistics for 192.168.1.13:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 4ms, Average = 1ms
C:\>
Laptop-PT Laptop2
Laptop-PT Laptop1
PC-PT client1
PC-PT client2

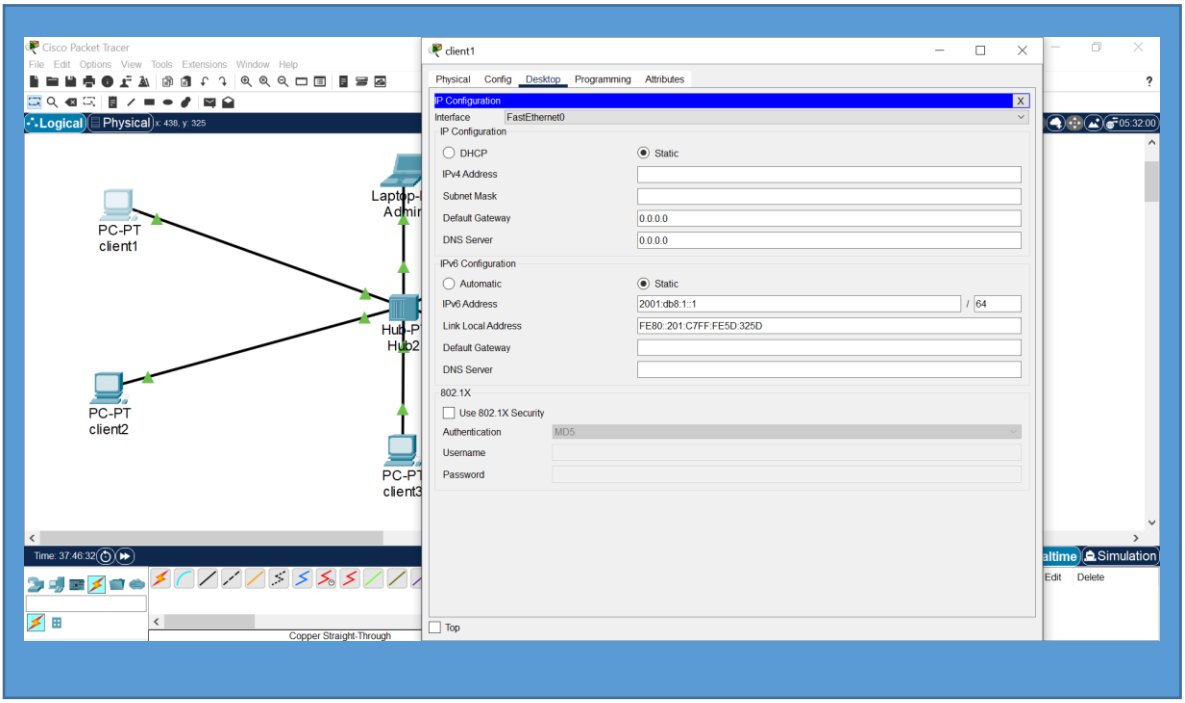
TÂCHE 4 : CONFIGURATION IPv6

Reproduire la topologie proposée et configurer les adresses IPv6 indiquées dans le tableau.

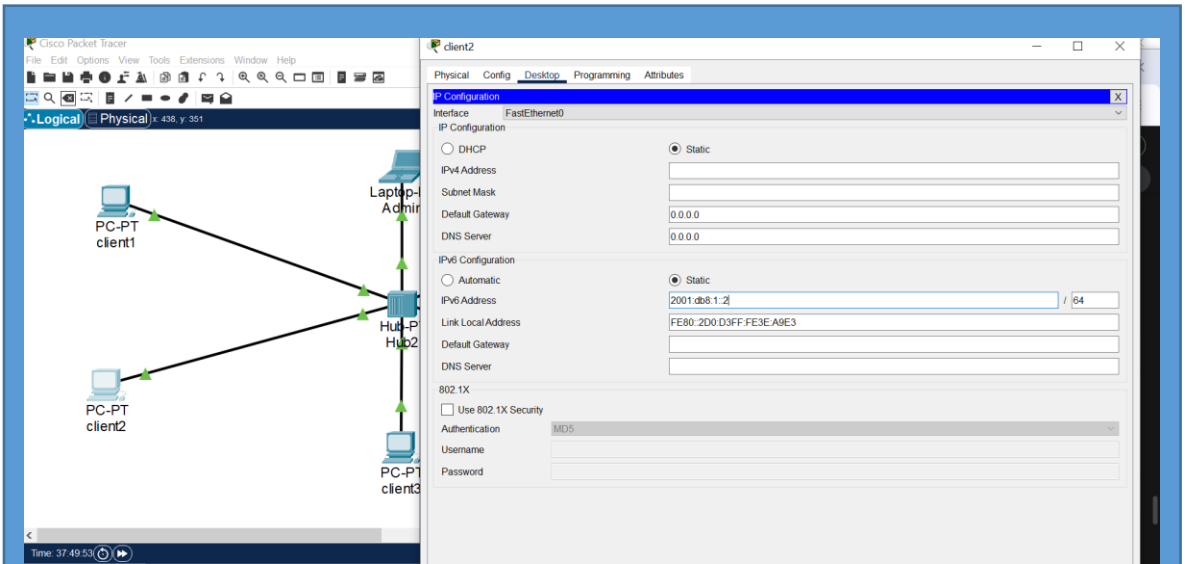
Reproduction de la topologie dans Cisco Packet Tracer



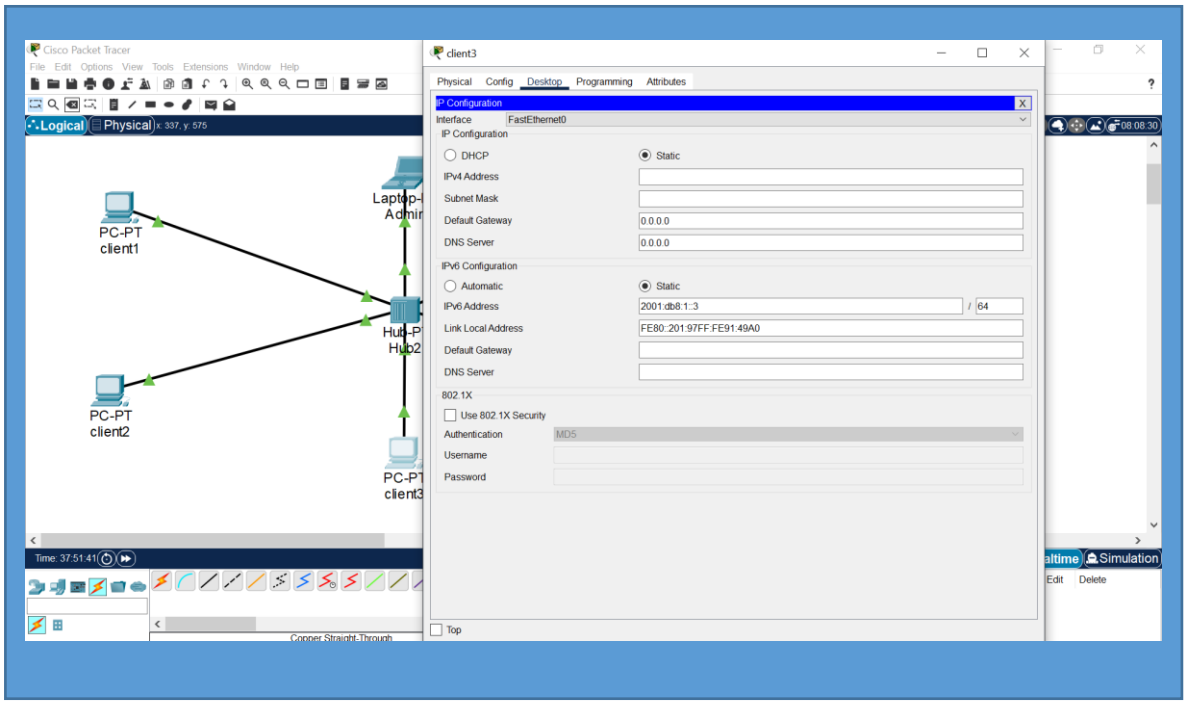
Configuration des adresses IPV6



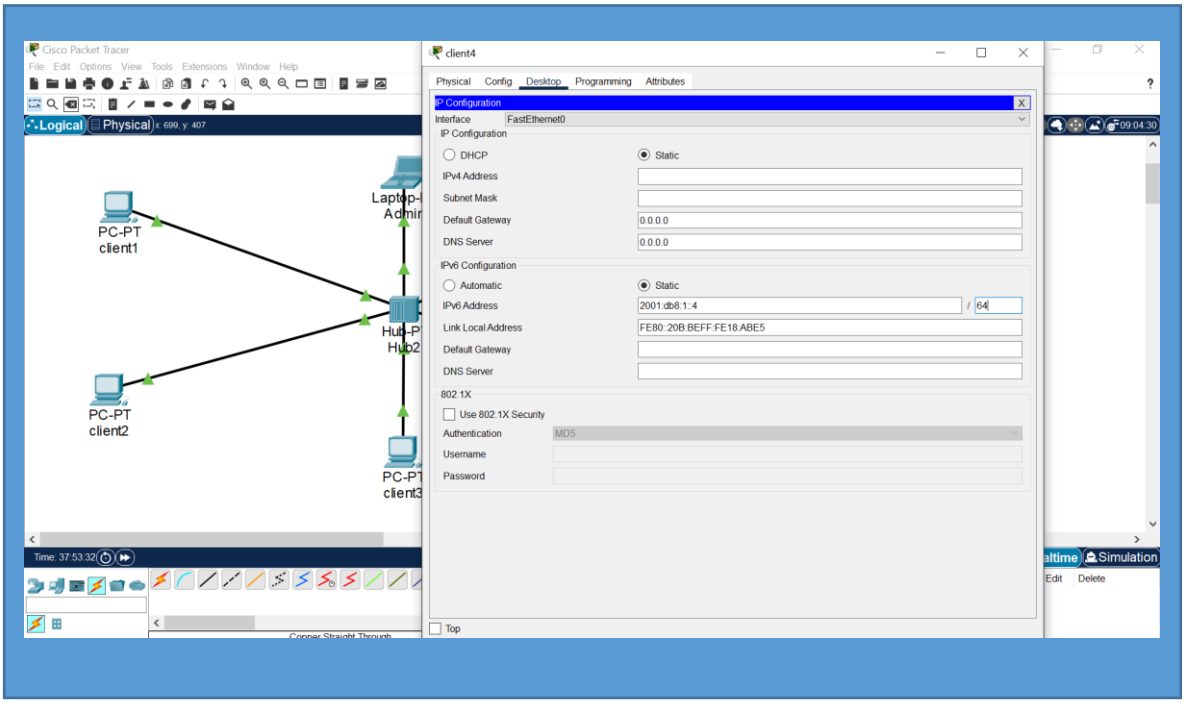
33.



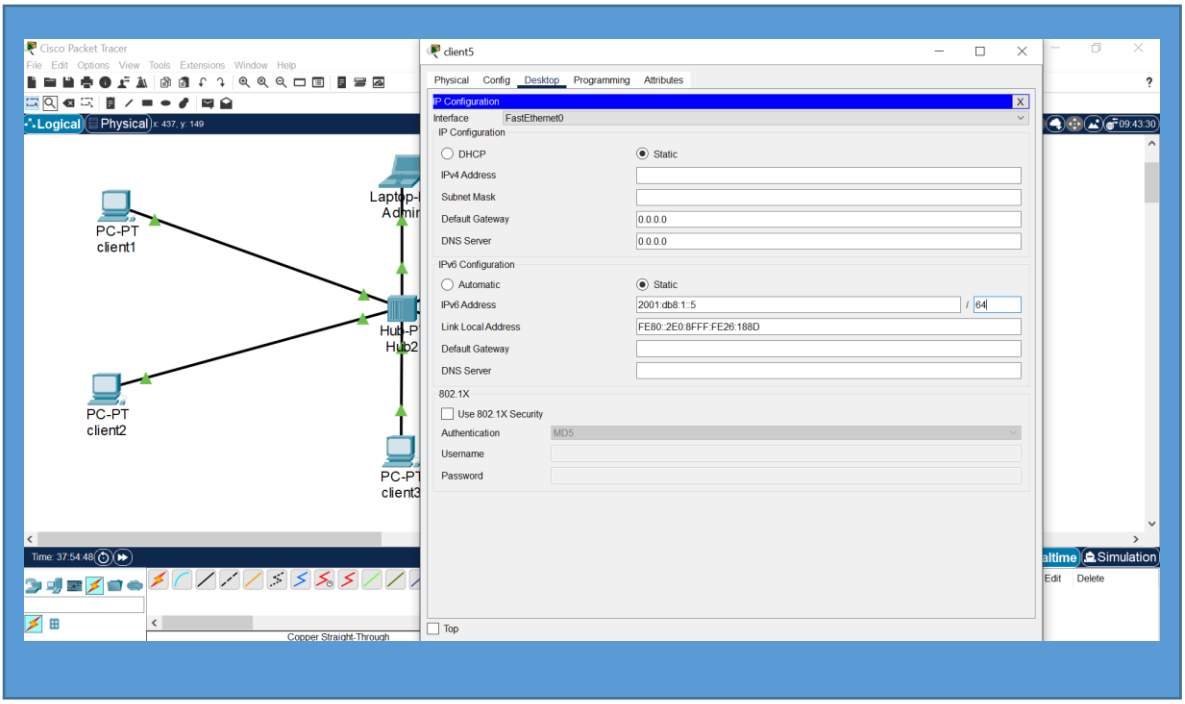
34.


Cisco Packet Tracer
client3
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
DHCP
Static
IPv4 Address
Subnet Mask
Default Gateway
0.0.0.0
DNS Server
IPv6 Configuration
Automatic
IPv6 Address
2001:db8:1::3
/ 64
Link Local Address
FE80::201:97FF:FE91:49A0
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top
PC-PT client1
PC-PT client2
PC-PT client3
Hub-PT Hub2
Logical
Physical
Time: 37:51:41
Simulation
Edit
Delete

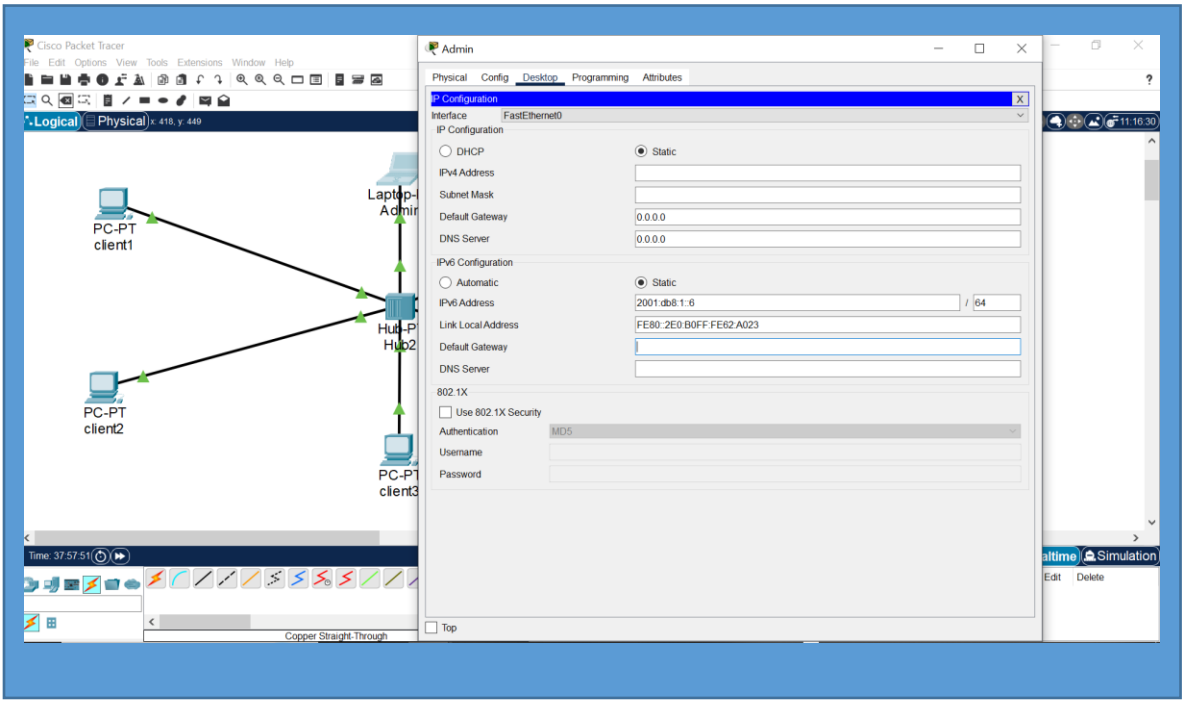
35.


Cisco Packet Tracer
client4
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
DHCP
Static
IPv4 Address
Subnet Mask
Default Gateway
0.0.0.0
DNS Server
IPv6 Configuration
Automatic
IPv6 Address
2001:db8:1::4
/ 64
Link Local Address
FE80::20B:BEFF:FE18:ABE5
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top
PC-PT client1
PC-PT client2
PC-PT client3
Hub-PT Hub2
Logical
Physical
Time: 37:53:32
Simulation
Edit
Delete

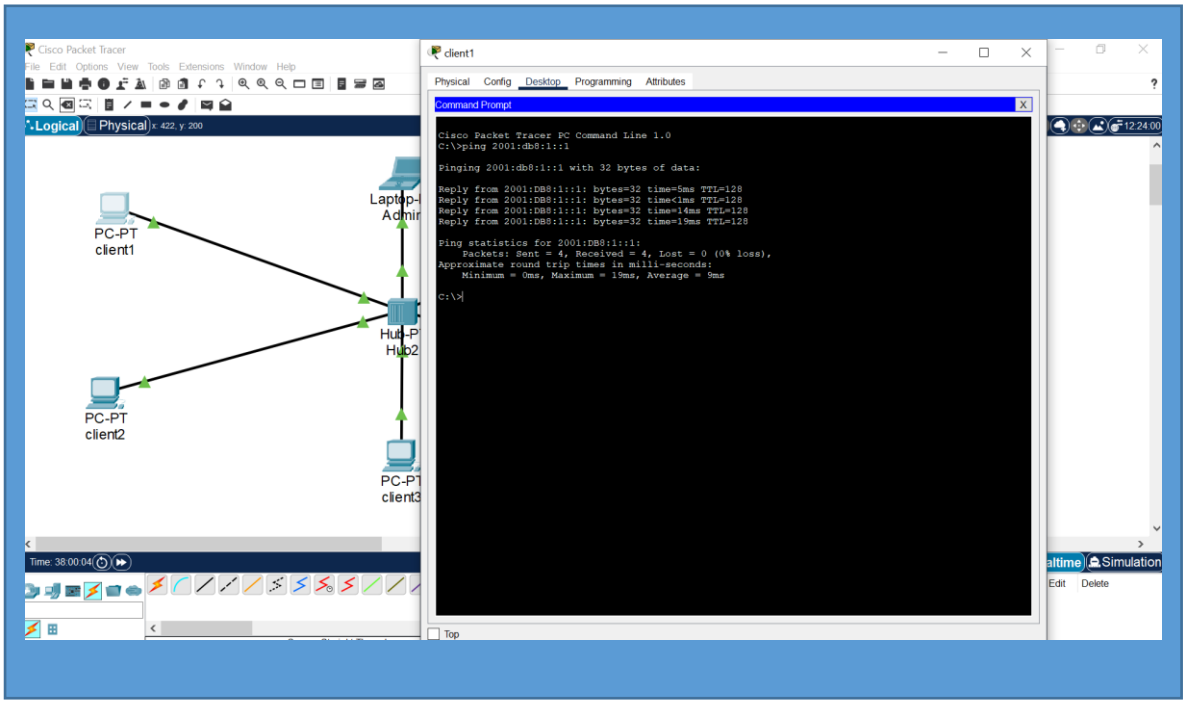
36.

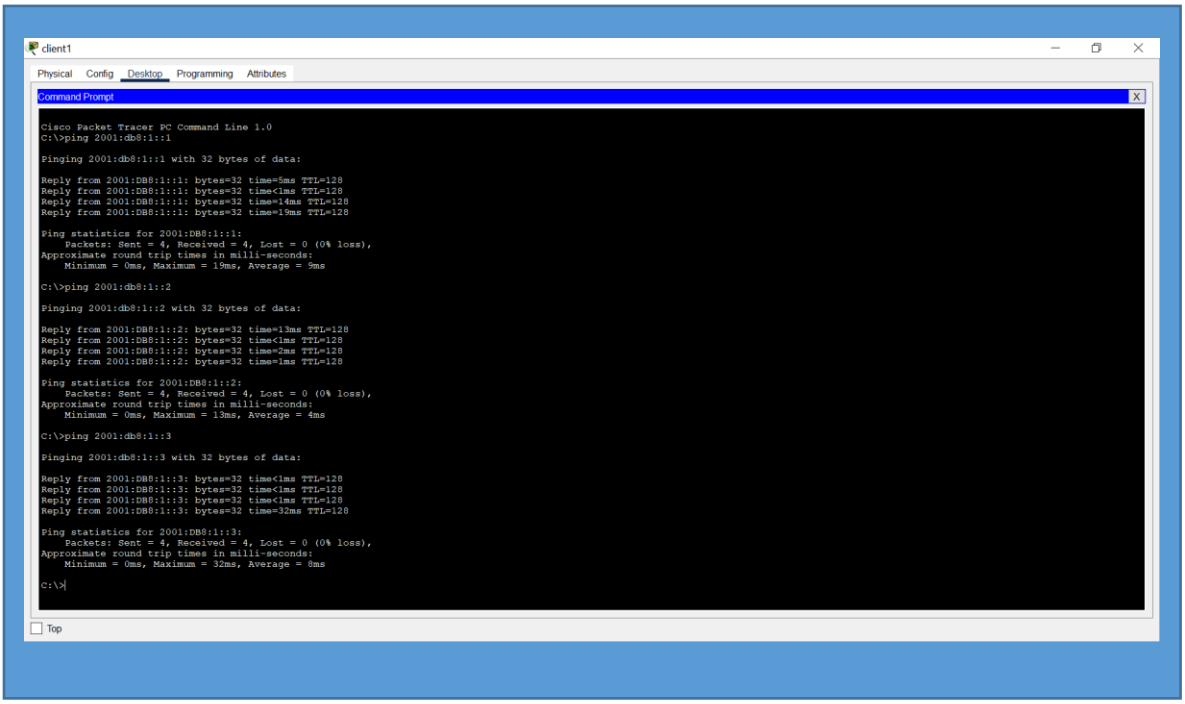

Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical Physical x: 437, y: 149
PC-PT client1
PC-PT client2
PC-PT client3
Hub-PT Hub2
Time: 37:54:48
Copper Straight-Through
client5
Physical Config Desktop Programming Attributes
IP Configuration
Interface FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
Subnet Mask
Default Gateway 0.0.0.0
DNS Server 0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address 2001:db8:1::5 / 64
Link Local Address FE80::2E0:8FFF:FE26:188D
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication MD5
Username
Password
Top
09:43:30
Simulation
Edit Delete

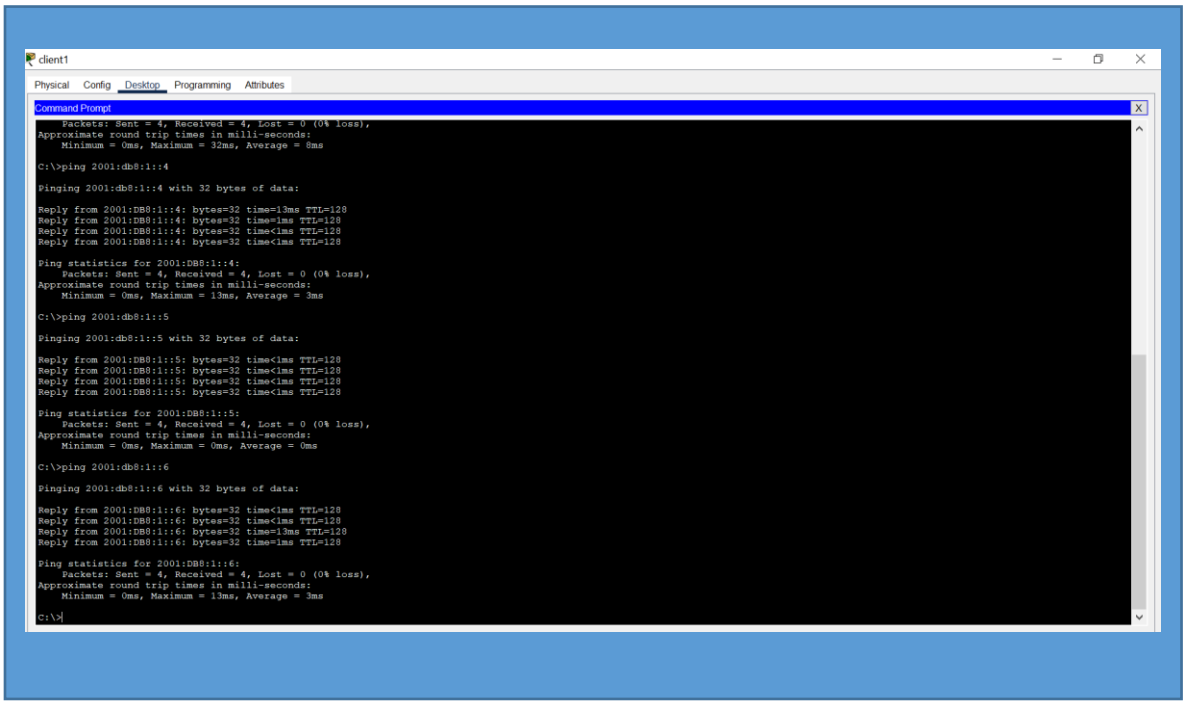
37.


Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical Physical x: 418, y: 449
PC-PT client1
PC-PT client2
PC-PT client3
Hub-PT Hub2
Time: 37:57:51
Copper Straight-Through
Admin
Physical Config Desktop Programming Attributes
IP Configuration
Interface FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
Subnet Mask
Default Gateway 0.0.0.0
DNS Server 0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address 2001:db8:1::6 / 64
Link Local Address FE80::2E0:B0FF:FE62:A023
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication MD5
Username
Password
Top
11:16:30
Simulation
Edit Delete

Tes de connectivite entre les equipements avec Ping IPV6



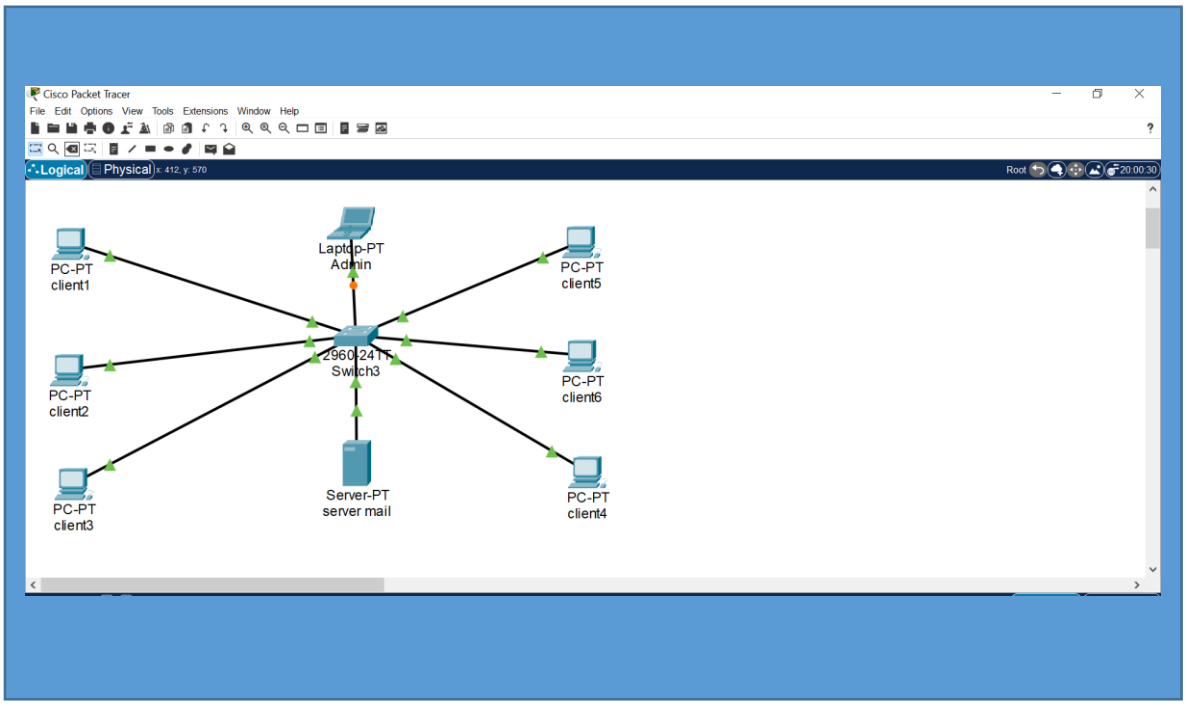




TÂCHE 5 : CONFIGURATION IPv6

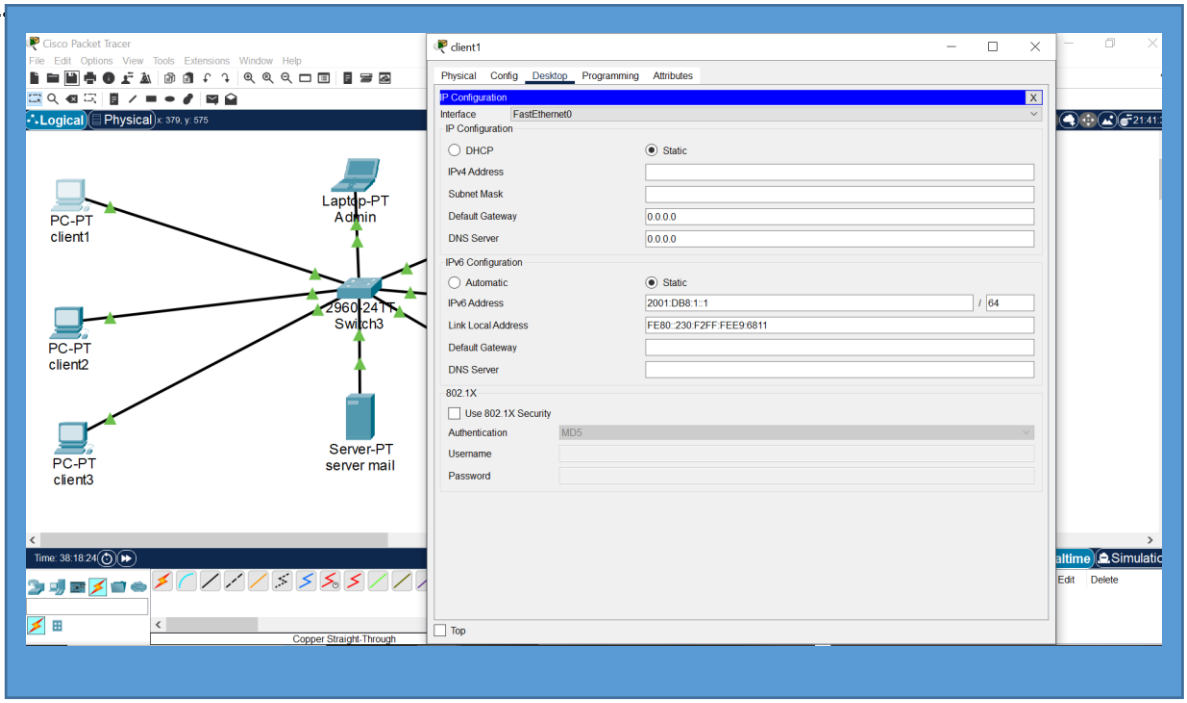
Reproduire la deuxième topologie IPv6 et attribuer les adresses correspondantes aux équipements.

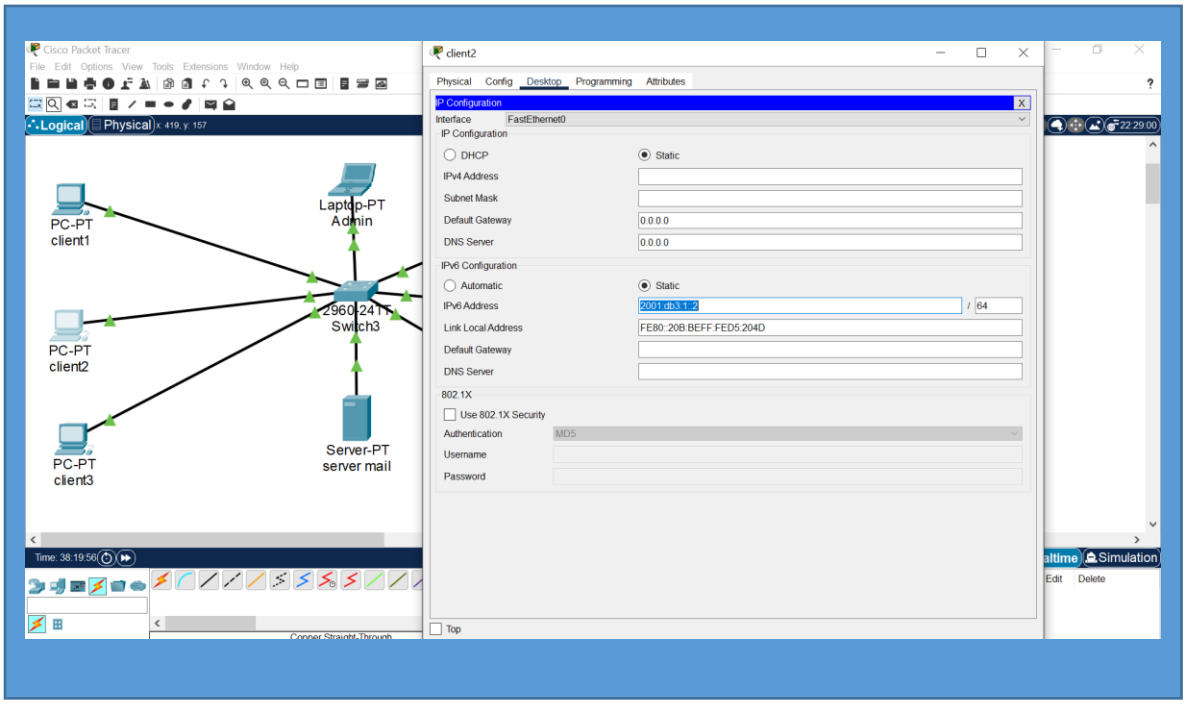
Creation de la topologie et connection des equipements

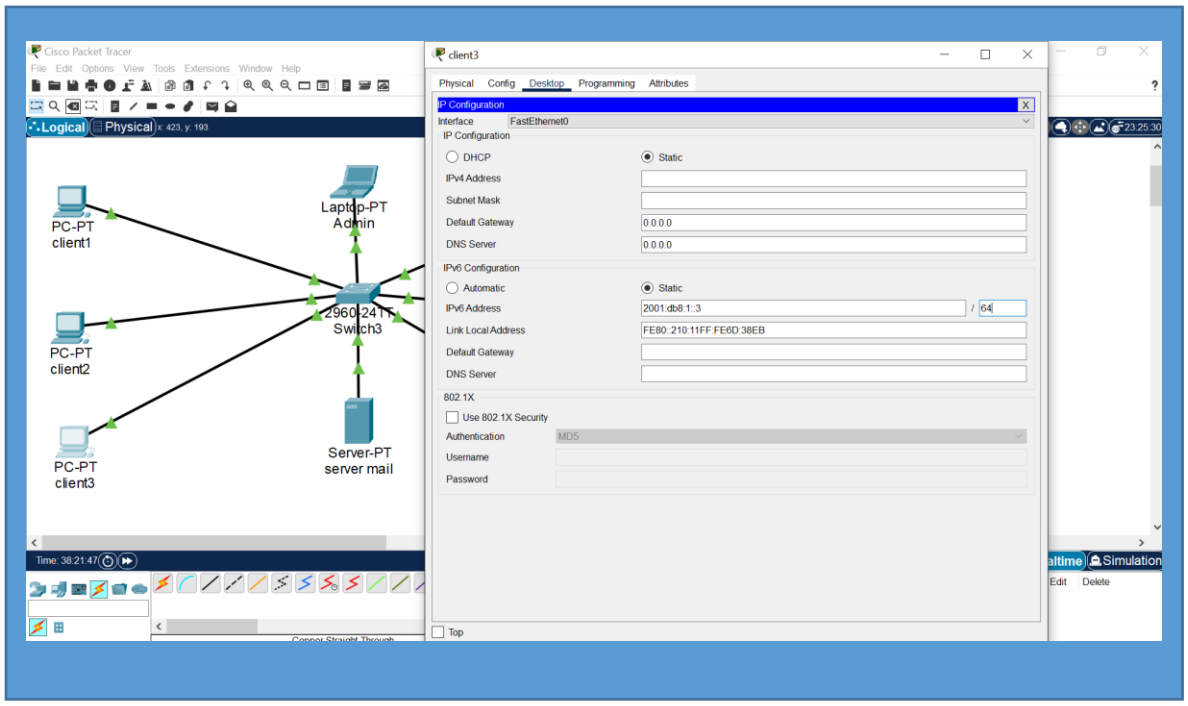


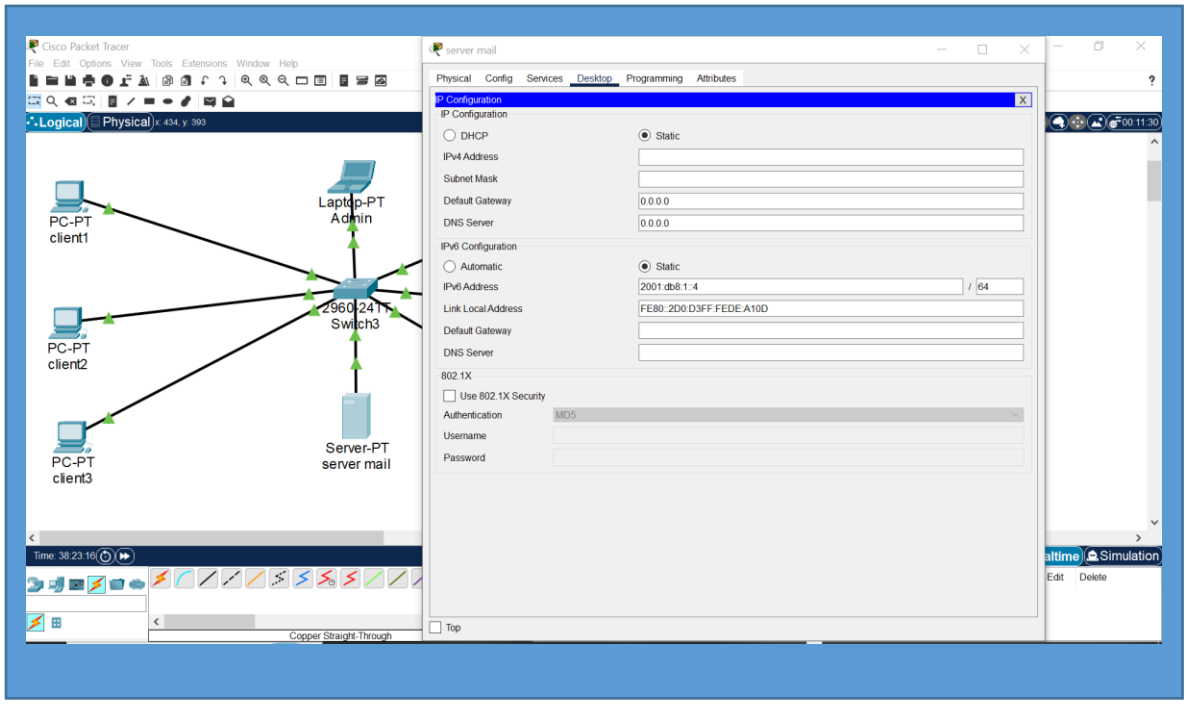
Attribution des adresses IPV6

42.

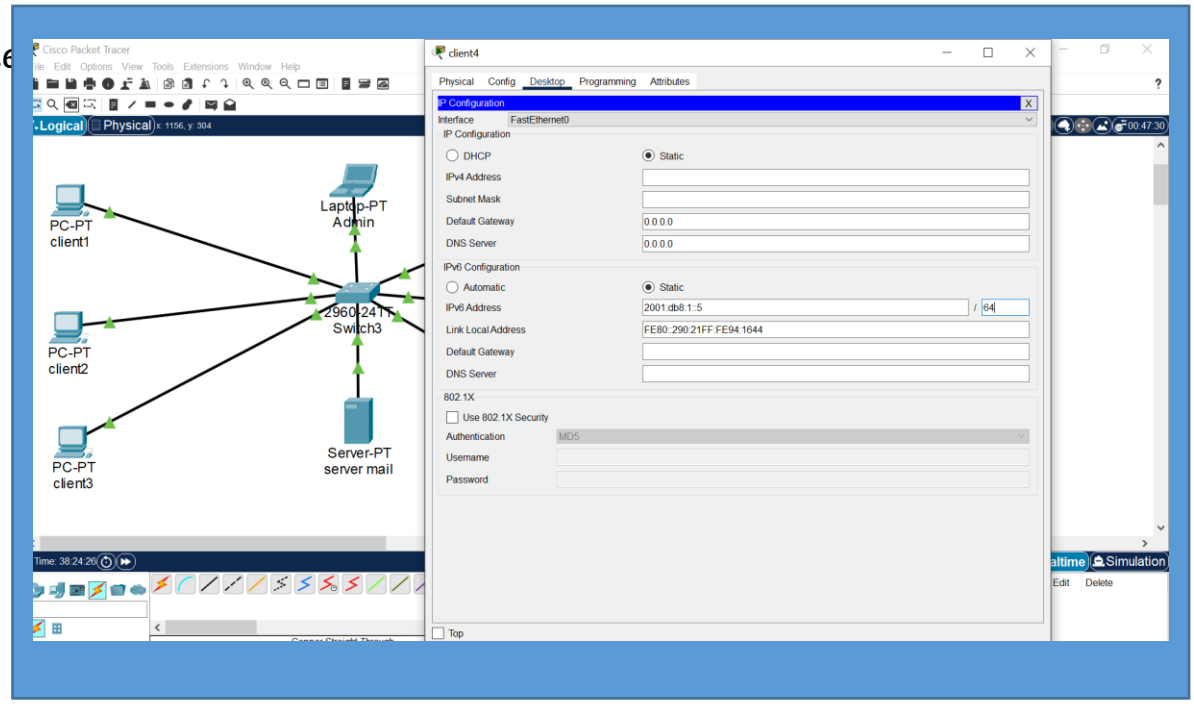


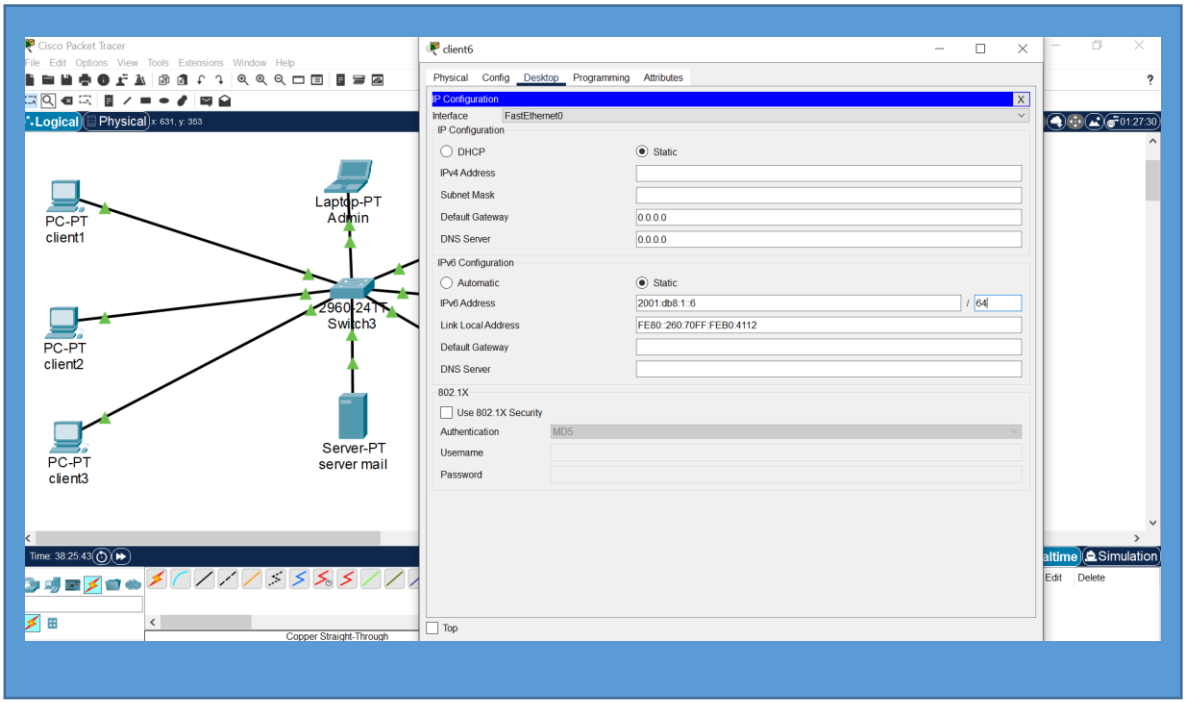


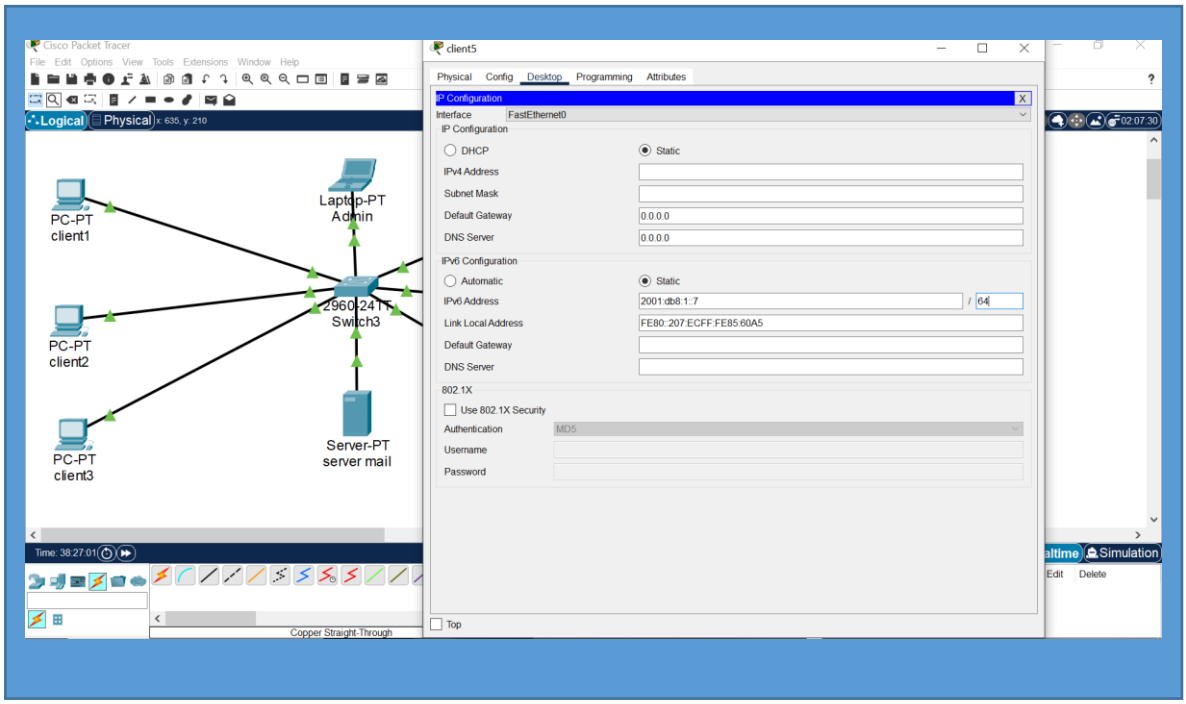

Cisco Packet Tracer
client3
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
DHCP
Static
IPv4 Address
Subnet Mask
Default Gateway
0.0.0.0
DNS Server
IPv6 Configuration
Automatic
IPv6 Address
2001:db8:1::3
/ 64
Link Local Address
FE80::210:11FF:FE6D:38EB
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top
Logical
Physical
PC-PT client1
PC-PT client2
PC-PT client3
Laptop-PT Admin
2960-24TT Switch3
Server-PT server mail
Time: 38:21:47
Edit
Delete
Simulation

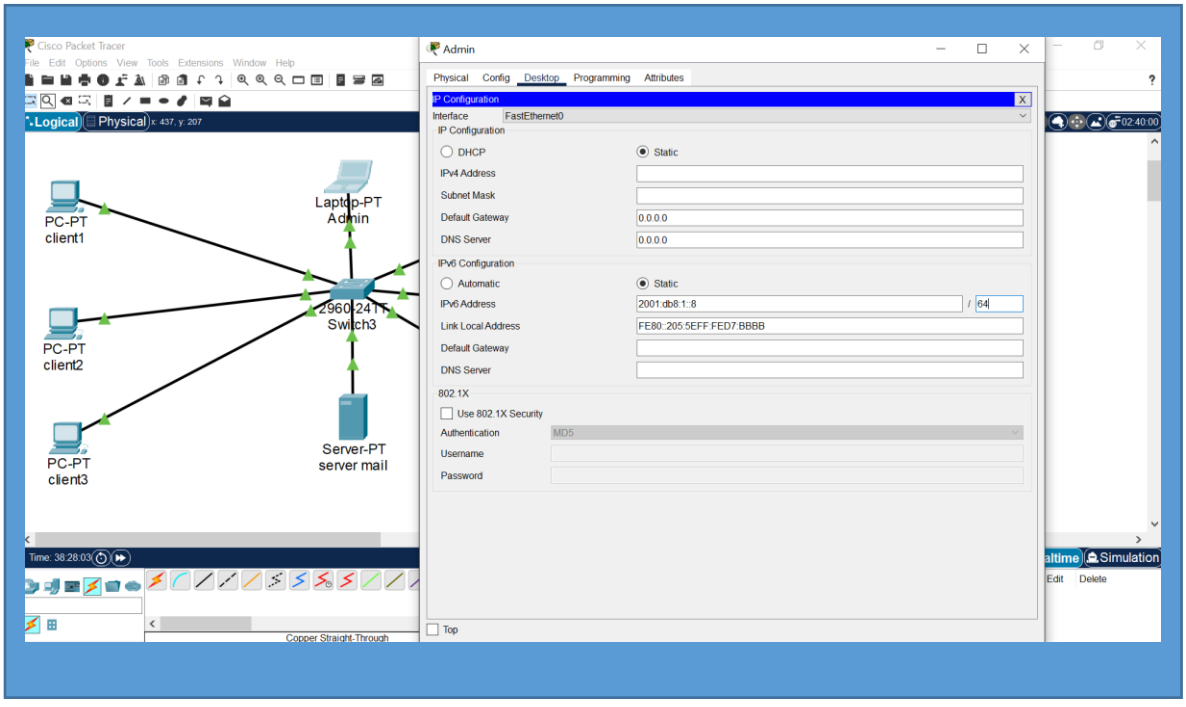

Cisco Packet Tracer
server mail
Physical
Config
Services
Desktop
Programming
Attributes
IP Configuration
DHCP
Static
IPv4 Address
Subnet Mask
Default Gateway
0.0.0.0
DNS Server
IPv6 Configuration
Automatic
IPv6 Address
2001:db8:1::4
/ 64
Link Local Address
FE80::2D0:D3FF:FEDE:A10D
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top
Logical
Physical
PC-PT client1
PC-PT client2
PC-PT client3
Laptop-PT Admin
2960-24TT Switch3
Server-PT server mail
Time: 38:23:16
Copper Straight-Through
Edit
Delete
Simulation

46

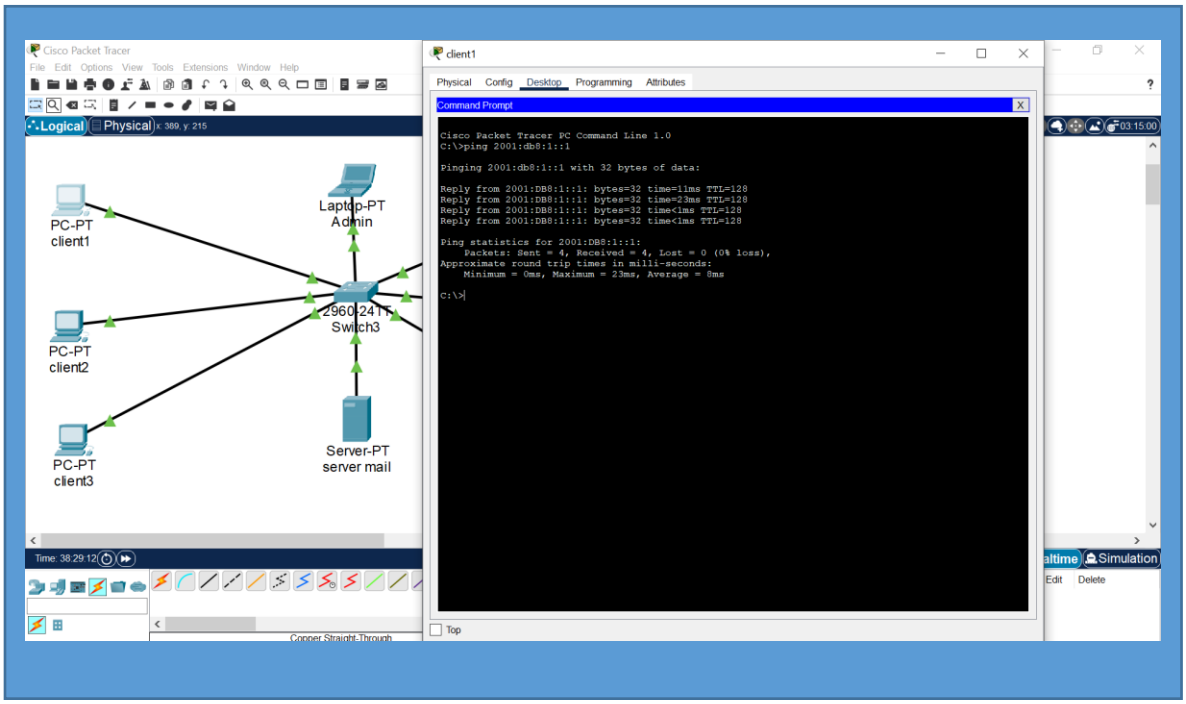

Cisco Packet Tracer
Logical
Physical
PC-PT client1
Laptop-PT Admin
2960-24TT Switch3
PC-PT client2
PC-PT client3
Server-PT server mail
client4
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
DHCP
Static
IPv4 Address
Subnet Mask
Default Gateway
0.0.0.0
DNS Server
IPv6 Configuration
Automatic
IPv6 Address
2001:db8:1::5
64
Link Local Address
FE80::290:21FF:FE94:1644
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top
Simulation
Edit
Delete

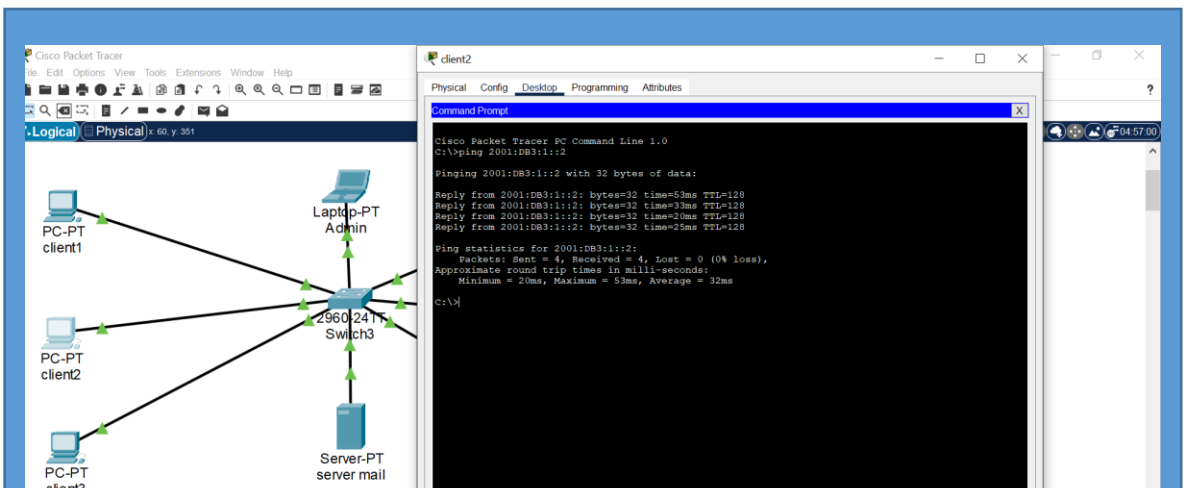

Cisco Packet Tracer
Logical
Physical
PC-PT client1
Laptop-PT Admin
2960-24TT Switch3
PC-PT client2
PC-PT client3
Server-PT server mail
client6
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
DHCP
Static
IPv4 Address
Subnet Mask
Default Gateway
0.0.0.0
DNS Server
IPv6 Configuration
Automatic
IPv6 Address
2001:db8:1::6
64
Link Local Address
FE80::260:70FF:FEB0:4112
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top
Copper Straight-Through
Simulation
Edit
Delete

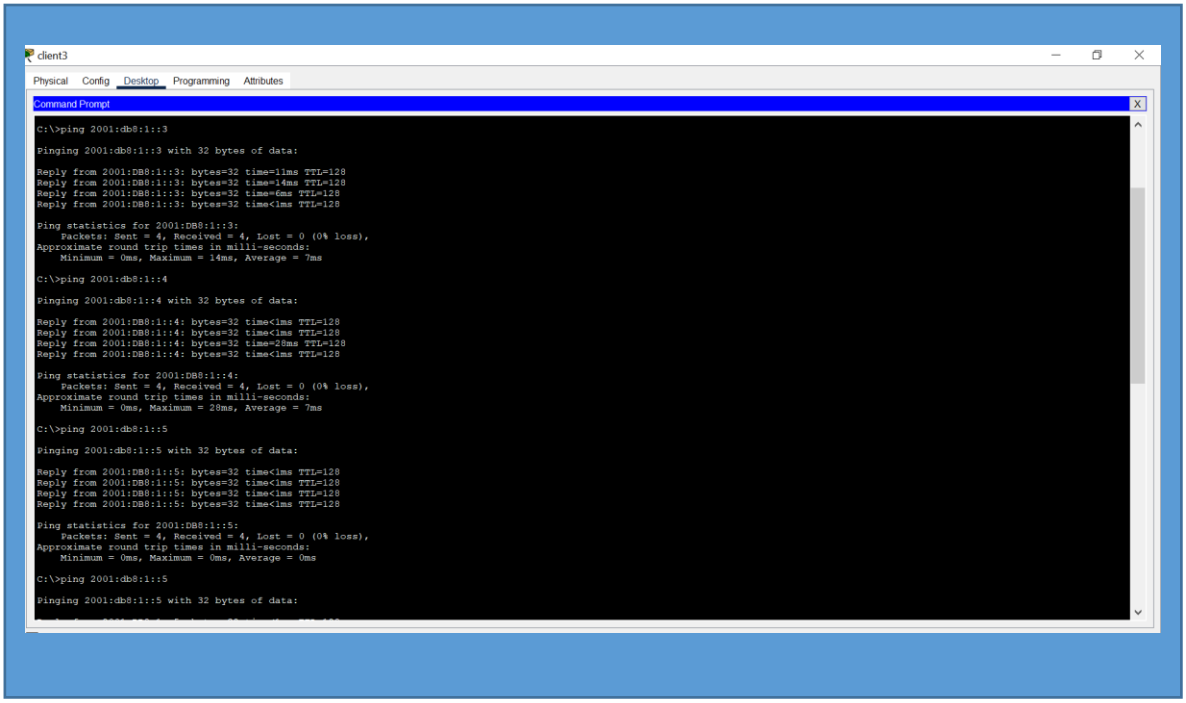

Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical Physical x: 635, y: 210
PC-PT client1
Laptop-PT Admin
2960-24TT Switch3
PC-PT client2
Server-PT server mail
PC-PT client3
Time: 38:27:01
Copper Straight-Through
client5
Physical Config Desktop Programming Attributes
IP Configuration
Interface FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
Subnet Mask
Default Gateway 0.0.0.0
DNS Server 0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address 2001:db8:1::7 / 64
Link Local Address FE80::207:ECFF:FE85:60A5
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication MD5
Username
Password
Top
02:07:30
Simulation
Edit Delete

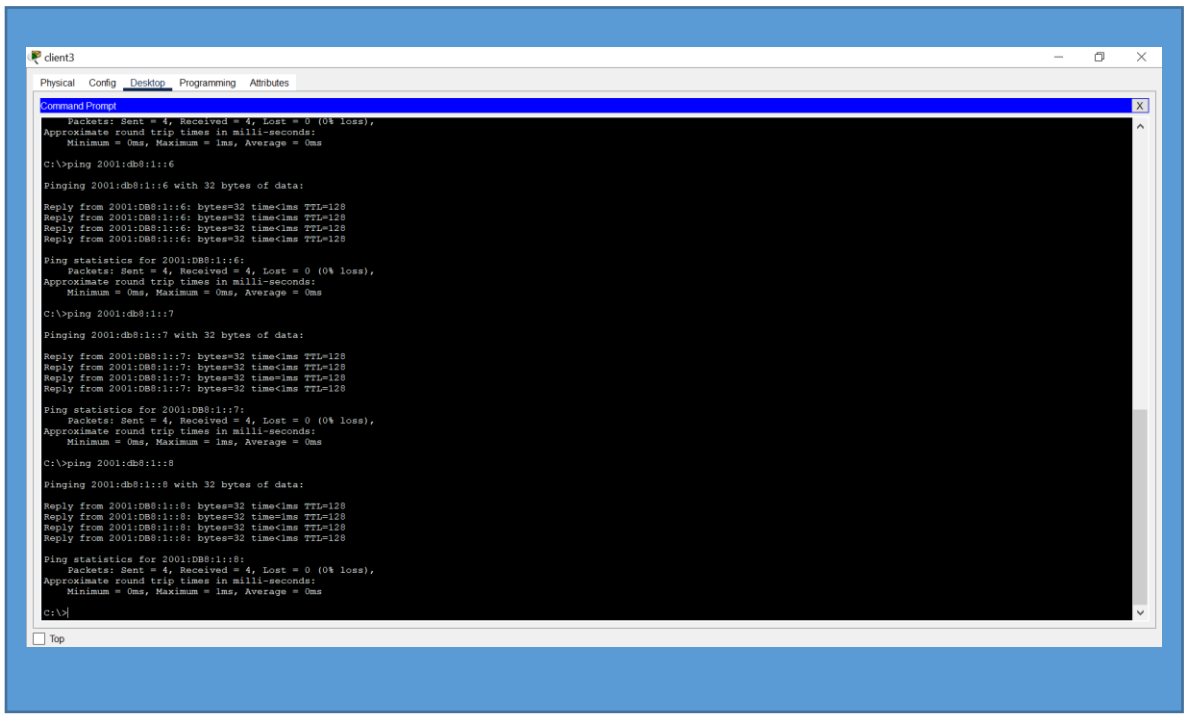

Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical Physical x: 437, y: 207
PC-PT client1
Laptop-PT Admin
2960-24TT Switch3
PC-PT client2
Server-PT server mail
PC-PT client3
Time: 38:28:03
Copper Straight-Through
Admin
Physical Config Desktop Programming Attributes
IP Configuration
Interface FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
Subnet Mask
Default Gateway 0.0.0.0
DNS Server 0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address 2001:db8:1::8 / 64
Link Local Address FE80::205:5EFF:FED7:BBBB
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication MD5
Username
Password
Top
02:40:00
Simulation
Edit Delete

Verification de la connectivite



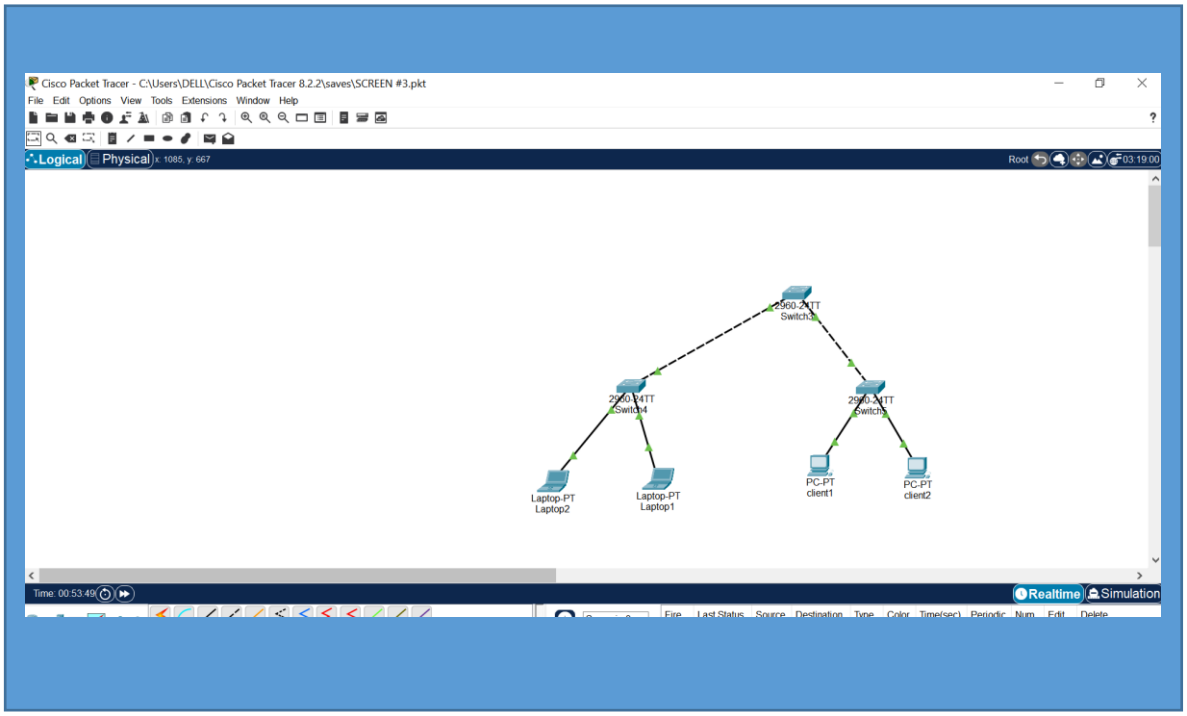



client3
Physical
Config
Desktop
Programming
Attributes
Command Prompt
C:\>ping 2001:db8:1::3
Pinging 2001:db8:1::3 with 32 bytes of data:
Reply from 2001:DB8:1::3: bytes=32 time=11ms TTL=128
Reply from 2001:DB8:1::3: bytes=32 time=14ms TTL=128
Reply from 2001:DB8:1::3: bytes=32 time=6ms TTL=128
Reply from 2001:DB8:1::3: bytes=32 time<1ms TTL=128
Ping statistics for 2001:DB8:1::3:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 14ms, Average = 7ms
C:\>ping 2001:db8:1::4
Pinging 2001:db8:1::4 with 32 bytes of data:
Reply from 2001:DB8:1::4: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::4: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::4: bytes=32 time=28ms TTL=128
Reply from 2001:DB8:1::4: bytes=32 time<1ms TTL=128
Ping statistics for 2001:DB8:1::4:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 28ms, Average = 7ms
C:\>ping 2001:db8:1::5
Pinging 2001:db8:1::5 with 32 bytes of data:
Reply from 2001:DB8:1::5: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::5: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::5: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::5: bytes=32 time<1ms TTL=128
Ping statistics for 2001:DB8:1::5:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 2001:db8:1::5
Pinging 2001:db8:1::5 with 32 bytes of data:

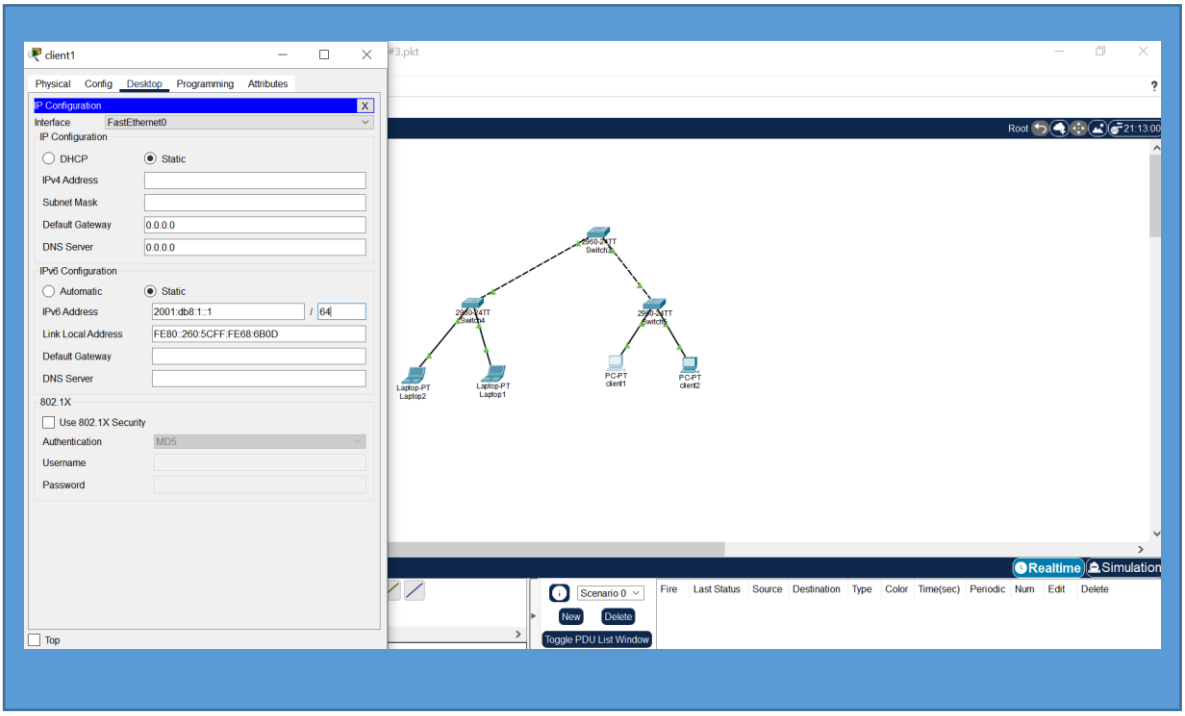

client3
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>ping 2001:db8:1::6
Pinging 2001:db8:1::6 with 32 bytes of data:
Reply from 2001:DB8:1::6: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::6: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::6: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::6: bytes=32 time<1ms TTL=128
Ping statistics for 2001:DB8:1::6:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 2001:db8:1::7
Pinging 2001:db8:1::7 with 32 bytes of data:
Reply from 2001:DB8:1::7: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::7: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::7: bytes=32 time=1ms TTL=128
Reply from 2001:DB8:1::7: bytes=32 time<1ms TTL=128
Ping statistics for 2001:DB8:1::7:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>ping 2001:db8:1::8
Pinging 2001:db8:1::8 with 32 bytes of data:
Reply from 2001:DB8:1::8: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::8: bytes=32 time=1ms TTL=128
Reply from 2001:DB8:1::8: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::8: bytes=32 time<1ms TTL=128
Ping statistics for 2001:DB8:1::8:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>
Top

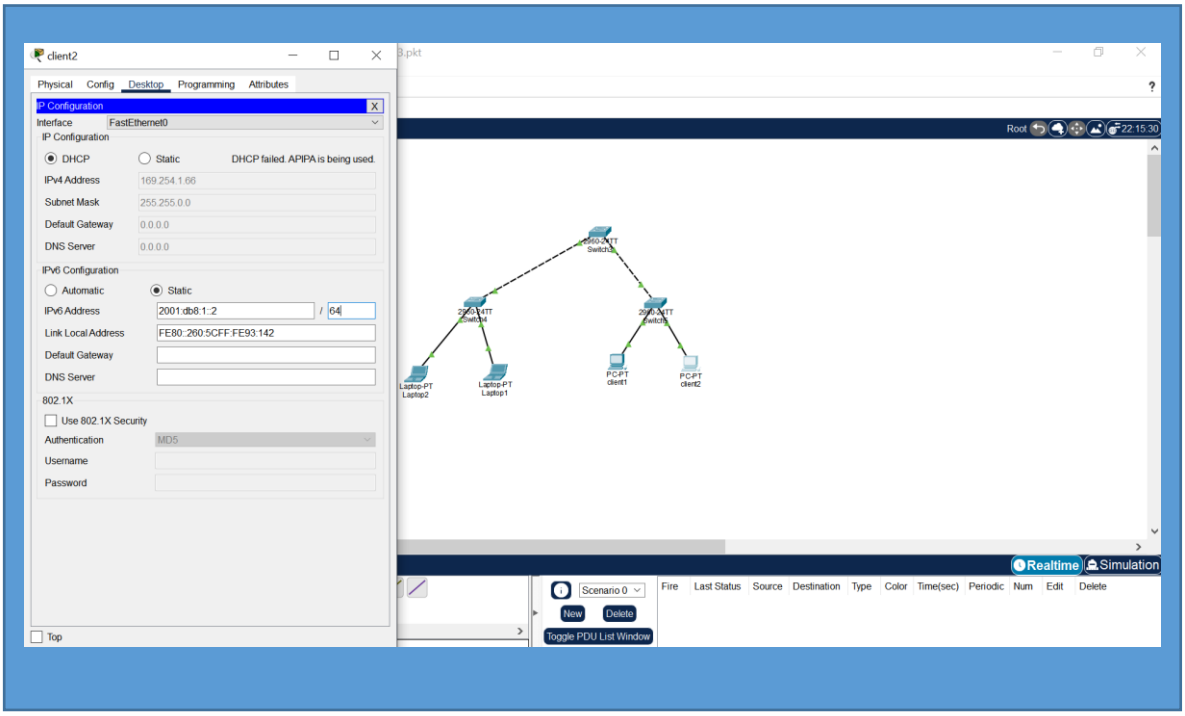
TÂCHE 6 : CONFIGURATION IPv6 Choisir une topologie parmi celles proposées, configurer les adresses IPv6 et tester la communication entre les équipements.

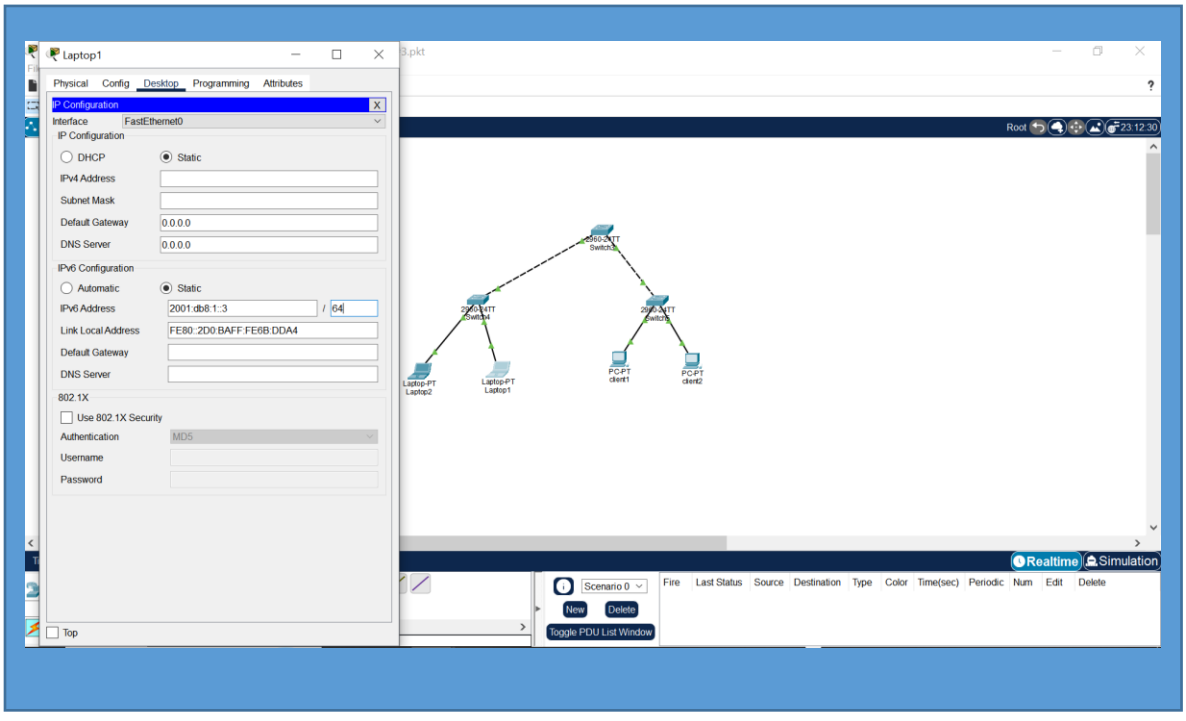
Choix de la topologie

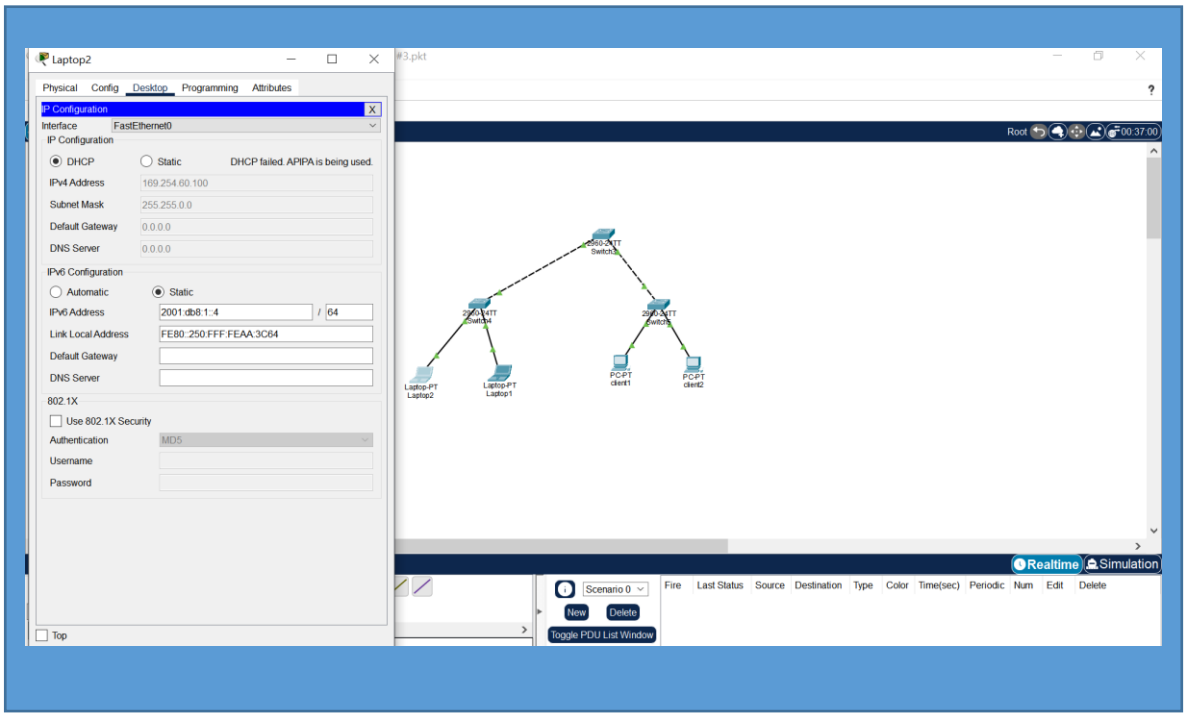


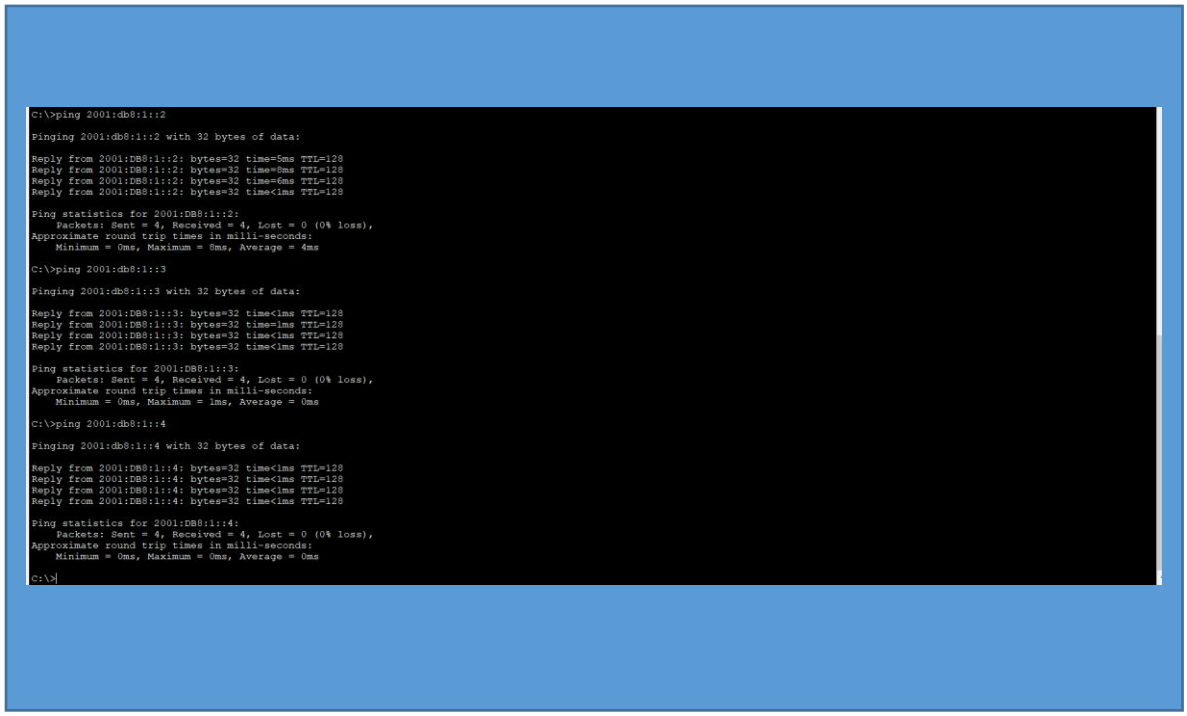
Configuration des adresses IPV6




client2
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
DHCP
Static
DHCP failed. APIPA is being used.
IPv4 Address
169.254.1.66
Subnet Mask
255.255.0.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
2001:db8:1::2
/ 64
Link Local Address
FE80::260:5CFF:FE93:142
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top
Root
Realtime
Simulation
Scenario 0
New
Delete
Toggle PDU List Window
Fire
Last Status
Source
Destination
Type
Color
Time(sec)
Periodic
Num
Edit
Delete


Laptop1
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
DHCP
Static
IPv4 Address
Subnet Mask
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
2001:db8:1::3
/ 64
Link Local Address
FE80::2D0:BAFF:FE6B:DDA4
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top
Root
Realtime
Simulation
Scenario 0
New
Delete
Toggle PDU List Window
Fire
Last Status
Source
Destination
Type
Color
Time(sec)
Periodic
Num
Edit
Delete


Laptop2
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
DHCP failed. APIPA is being used.
IPv4 Address
169.254.60.100
Subnet Mask
255.255.0.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
2001:db8:1::4
/ 64
Link Local Address
FE80::250:FFF:FEAA:3C64
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top
#3.pkt
Root
00:37:00
Switch3
Switch4
Switch5
Laptop-PT
Laptop2
Laptop-PT
Laptop1
PC-PT
client1
PC-PT
client2
Realtime
Simulation
Scenario 0
New
Delete
Toggle PDU List Window
Fire
Last Status
Source
Destination
Type
Color
Time(sec)
Periodic
Num
Edit
Delete


C:\>ping 2001:db8:1::2
Pinging 2001:db8:1::2 with 32 bytes of data:
Reply from 2001:DB8:1::2: bytes=32 time=5ms TTL=128
Reply from 2001:DB8:1::2: bytes=32 time=8ms TTL=128
Reply from 2001:DB8:1::2: bytes=32 time=6ms TTL=128
Reply from 2001:DB8:1::2: bytes=32 time<1ms TTL=128
Ping statistics for 2001:DB8:1::2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 8ms, Average = 4ms
C:\>ping 2001:db8:1::3
Pinging 2001:db8:1::3 with 32 bytes of data:
Reply from 2001:DB8:1::3: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::3: bytes=32 time=1ms TTL=128
Reply from 2001:DB8:1::3: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::3: bytes=32 time<1ms TTL=128
Ping statistics for 2001:DB8:1::3:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>ping 2001:db8:1::4
Pinging 2001:db8:1::4 with 32 bytes of data:
Reply from 2001:DB8:1::4: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::4: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::4: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::4: bytes=32 time<1ms TTL=128
Ping statistics for 2001:DB8:1::4:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>

CONCLUSION GÉNÉRALE

Au terme de ce troisième travail dirigé, nous avons approfondi nos connaissances sur l'adressage IPv4 et IPv6 à travers le logiciel Cisco Packet Tracer.

Nous avons appris à reproduire différentes topologies, à attribuer des adresses IP aux équipements et à vérifier la communication entre les machines à l'aide des commandes Ping.

Ce travail nous a permis de mieux comprendre le fonctionnement de l'adressage réseau et de développer nos compétences pratiques en configuration des réseaux informatiques.

Le TD a été réalisé à travers les différentes tâches de configuration et de vérification demandées.

